Aphanics at the Pathion Level: Spectrum Stratification and the Universal Clifford Triad

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Abstract

We extend the Aphanes framework from the sedenion algebra $\mathbb{S}$ to its Cayley–Dickson double, the pathion algebra $\mathbb{P} = \mathbb{S} \oplus \mathbb{S} \cdot \ell$ of dimension 32. At sedenion level, prior work establishes that for any aphanic basepoint p the closed-form Berry curvature operator $\mathcal{C}_{a,b}(p)$ has a four-dimensional kernel structure giving $\mathcal{F}(p) \cong \mathfrak{u}(1) \oplus \mathfrak{su}(2)$ and a G_2-equivariant rank-4 closure. Lifting these results to pathion exposes structurally new phenomena: the Gram operator $G_p = L_p^\top L_p$ has spectrum $\{0, \|p\|^2, 2\|p\|^2\}$ at dyadic basepoints and an eight-valued spectrum at generic aphanics. We establish two theorems describing the kill set of $a \mapsto \mathcal{C}_{a,*}(p)$; in each, the containment (sufficiency) direction is proved symbolically, while the exact kill-set dimension (necessity) is certified numerically (Appendix A). The *Pathion Universal Kill Set Theorem* states that at every pathion-aphanic p with $\dim K_p = 4$, this kill set is exactly the three-dimensional space $\mathfrak{q}_p^{\text{univ}} = \text{span}_{\mathbb{R}}\{p, \ell_{16}, \ell_{16} \cdot p\}$. The *Pathion Q-Centralizer Vanishing Theorem at Dyadic Basepoint* states that under the additional hypothesis of a three-valued Gram spectrum of canonical Stratum-II type, the kill set extends to a five-dimensional space $\mathfrak{q}_p^{(32)} \supset \mathfrak{q}_p^{\text{univ}}$. Both theorems follow from a structural result which is the heart of this paper: the three operators $L_p, L_{\ell_{16}}, L_{\ell_{16} \cdot p}$ form a pairwise anti-commuting Clifford triad on $\mathbb{P}$. We prove this Triad Theorem algebraically, reducing the third pair to a sedenion identity $\Delta(s_1, s_2) = 0$ for orthogonal imaginary $s_1, s_2 \in \text{Im}(\mathbb{S})$, which in turn reduces by a four-block Cayley–Dickson decomposition to a three-variable octonion identity $\Omega(A, B; c) = 0$. The latter falls to a two-line argument from polarization, Artin's theorem on two-generator subalgebras of the octonions, and the composition norm form. The dim-5 structure $\mathfrak{q}_p^{(32)} = \mathfrak{q}_p^{\text{univ}} \oplus \mathfrak{q}_p^{\text{anti}}$ is shown to be $\mathbb{Z}_2$-graded: each summand is independently Clifford-internal at every aphanic basepoint, but mixed pairs across the grading do not anti-commute. Hence the dyadic-only extension to $\mathfrak{q}_p^{(32)}$ uses a mechanism distinct from the universal Clifford triad. All symbolically proved identities are additionally certified numerically; each certification states its epistemic basis in Appendix A.

1 Introduction

The Aphanes framework studies a class of algebraically hidden generators in non-associative Cayley–Dickson algebras above the octonions. The core equation $p \cdot \mathcal{A}(p) = 0$ with $\mathcal{A}(p) \neq 0$ captures a structurally real generator annihilated by its own system's left-multiplication. At sedenion level $\mathbb{S}$ (dimension 16), prior work established the algebraic characterization of aphanic basepoints and the universal $4-8-4$ spectrum of the Gram operator [1], and a closed-form Berry curvature operator [2] $\mathcal{C}_{a,b}(p)$ whose four-dimensional kernel space carries a $\mathfrak{u}(1) \oplus \mathfrak{su}(2)$ Lie structure with global holonomy group $U(2) = (U(1) \times SU(2))/\mathbb{Z}_2$, the gauge group of the electroweak interaction, arrived at from pure sedenion algebra without physics input.

A natural question is whether these structures persist at higher levels of the Cayley–Dickson tower. The pathion algebra $\mathbb{P} = \mathbb{S} \oplus \mathbb{S} \cdot \ell_{16}$ of dimension 32 provides the first non-trivial test. We establish that the framework lifts in a refined form (sufficiency directions symbolically, exact

kill-set dimensions by numerical certification): the rank-4 kernel persists, but the universal kill set of the curvature operator drops from dimension 5 at sedenion to dimension 3 at pathion, with a dimension-5 extension surviving only at basepoints of three-valued spectrum.

The structural heart of these results is a Clifford-like triad of three pairwise anti-commuting operators canonically attached to each aphanic basepoint. We prove the triad theorem algebraically by reducing through a Cayley–Dickson tower: the pathion identity reduces to a sedenion identity, which reduces to an octonion identity, which falls to Artin's theorem on two-generator subalgebras. The chain hides the non-associative complexity by trapping it in the well-developed alternative-algebra structure of the octonions.

Organization. Section 2 fixes notation for the Cayley–Dickson recursion, the aphanic locus, and the Berry curvature operator. Section 3 describes the pathion spectrum stratification. Section 4 states and proves the two pathion Q-centralizer vanishing theorems. Section 5 proves the Pathion Clifford Triad Theorem and unwinds the supporting algebraic chain (Lemma A, Delta-reduction, Omega-theorem). Section 6 describes the $\mathbb{Z}_2$-graded Clifford sublattice on $\mathfrak{q}_p^{(32)}$ that separates the universal from the dyadic-only kill set. Section 7 proves the Two-Regime Dyadic Closure Theorem underlying the dyadic extension. Section 8 discusses connections to the sedenion U(2) holonomy and open questions. Appendix A reports machine-precision certification of the key identities.

Main results. The substantive contributions of this paper are:

(1) **Theorem 4.1** (Pathion Universal Kill Set): for every pathion-aphanic p with $\dim K_p = 4$, the kill set of $a \mapsto \mathcal{C}_{a,*}(p)$ is the three-dimensional space $\mathfrak{q}_p^{\text{univ}} = \text{span}_{\mathbb{R}}\{p, \ell_{16}, \ell_{16} \cdot p\}$.

(2) **Theorem 4.3** (Pathion Dyadic Q-Centralizer Vanishing): at pathion-aphanic p with $\text{Spec}(G_p) = \{0, \|p\|^2, 2\|p\|^2\}$, the kill set extends to a five-dimensional space $\mathfrak{q}_p^{(32)} = \iota_1(V_p) \oplus \mathbb{R}\ell_{16} \oplus \iota_2(V_p)$.

(3) **Theorem 5.1** (Pathion Clifford Triad): the three operators $L_p, L_{\ell_{16}}, L_{\ell_{16}\cdot p}$ on $\mathbb{P}$ are skew-symmetric, mutually orthogonal, and pairwise anti-commute.

(4) **Theorem 5.7** (Octonion Ω Identity): for all $A, B, c \in \mathbb{O}$, $\bar{A}(Bc) + \bar{B}(Ac) - (c\bar{B})A - (c\bar{A})B = 0$.

(5) **Theorem 6.3** ($\mathbb{Z}_2$-Graded Clifford Sublattice): the dim-5 space $\mathfrak{q}_p^{(32)}$ decomposes as $\mathfrak{q}_p^{\text{univ}} \oplus \mathfrak{q}_p^{\text{anti}}$ where each summand is independently Clifford-internal but pairs across the grading are not.

(6) **Theorem 7.17** (Two-Regime Dyadic Closure): $[L_a, G_p] = 0$ for $a \in \mathfrak{q}_p^{\text{anti}}$ at every dyadic basepoint with three-valued Gram spectrum, by distinct mechanisms at canonical Stratum-II and at CD-aligned.

2 Preliminaries

2.1 Cayley–Dickson recursion

We use the canonical recursive Cayley–Dickson product. For a normed $*$-algebra A, its double $A^{(2)} = A \oplus A$ has product

$$(a, b)(c, d) = (ac - \bar{d}\, b,\ d\, a + b\, \bar{c}) \tag{1}$$

and conjugation $(a, b)^* = (\bar{a}, -b)$. Iterating from $\mathbb{R}$ produces $\mathbb{R} \to \mathbb{C} \to \mathbb{H} \to \mathbb{O} \to \mathbb{S} \to \mathbb{P} \to \cdots$ with dimensions $1, 2, 4, 8, 16, 32$. We write $\mathbb{O}$ for the octonions, $\mathbb{S}$ for the sedenions, and $\mathbb{P}$ for the pathions.

The norm form $\|x\|^2 = x\bar{x} = \bar{x}x \in \mathbb{R}$ holds in every Cayley–Dickson algebra. The composition law $\|xy\| = \|x\|\|y\|$ holds in $\mathbb{O}$ but fails in $\mathbb{S}$. Octonions are alternative; sedenions are flexible but not alternative; this exact transition is the source of the structural phenomena studied here.

For $p = (x, y) \in \mathbb{P}$ with $x, y \in \mathbb{S}$, write $\iota_1(x) = (x, 0)$ and $\iota_2(y) = (0, y)$, so $p = \iota_1(x) + \iota_2(y)$. The left-multiplication operator L_p acts on $\mathbb{P}$ by $L_p v = pv$, and similarly $R_p v = vp$. We write $L_p^{(\mathbb{S})}$ when the underlying algebra is $\mathbb{S}$, and L_p unambiguously when context fixes the algebra. We use ℓ_{16} for the unit doubling generator of $\mathbb{P}$ over $\mathbb{S}$, so $\ell_{16} = (0, 1)$ and $\ell_{16}^2 = -1$.

2.2 The aphanic locus

Definition 2.1. A point $p \in \mathbb{P}$ is *aphanic* if $\dim_{\mathbb{R}} \ker L_p > 0$. We write $\mathcal{P}^{\text{aph}} \subset \text{Im}(\mathbb{P})$ for the locus of imaginary aphanic points.

The sedenion-level characterization of [1] states that $p = \iota_1(x) + \iota_2(y) \in \mathbb{S}$ is aphanic if and only if $x, y \in \text{Im}(\mathbb{O})$, $\|x\| = \|y\|$, and $\langle x, y\rangle = 0$. At pathion the analogous characterization holds in two strata:

Proposition 2.2 (Pathion Aphanic Stratification)**.** *For $p = e_i + e_j$ with $i \in \{1, \ldots, 15\}$ and $j \in \{17, \ldots, 31\}$ (so the pathion slot i encodes a sedenion direction s_1 and slot $j - 16$ encodes a sedenion direction s_2), p is aphanic if and only if $s_1 \neq s_2$, and:*

- *if $\langle s_1, s_2\rangle_{\mathbb{S}} = 0$ in $\mathbb{R}^{15}$ but the pair lies in a common octonion sub-copy (*Stratum I*), then* $\dim \ker L_p = 12$ *and* $\text{Spec}(G_p) = \{0^{12}, 2^8, 4^{12}\}$*;*
- *otherwise (*Stratum II*),* $\dim \ker L_p = 4$ *and* $\text{Spec}(G_p) = \{0^4, 2^{24}, 4^4\}$*, the sedenion-recursive analog.*

In what follows, "aphanic" will refer to the generic Stratum II case $\dim K_p = 4$. The structurally important spectral fact is that this spectrum has only three distinct values: $0, \|p\|^2 = 2$, and $2\|p\|^2 = 4$. At generic aphanic basepoints (random imaginary halves with the orthogonality and equal-norm constraints), the spectrum has eight distinct values $\{0^4, \lambda_1^4, \ldots, \lambda_7^4\}$ with all multiplicities equal to four, summing to $8 \cdot 4 = 32 = \dim \mathbb{P}$.

2.3 Berry curvature operator and kill set

The closed-form Berry curvature operator [2, Theorem 3.7] attached to an aphanic basepoint p is

$$\mathcal{C}_{a,b}(p) = \pi_p\big(L_b\, G_p^+\, L_a - L_a\, G_p^+\, L_b\big)\pi_p \tag{2}$$

where $G_p = L_p^\top L_p$, G_p^+ is the Moore–Penrose pseudoinverse, and $\pi_p = I - G_p G_p^+$ is the orthogonal projector onto $\ker L_p$. The operator depends bilinearly on $a, b \in \text{Im}(\mathbb{P})$.

Definition 2.3 (Kill Set)**.** The *kill set* of the curvature operator at p is

$$\mathcal{K}(p) := \big\{\, a \in \text{Im}(\mathbb{P}) \;:\; \mathcal{C}_{a,b}(p) = 0 \text{ for all } b \in \text{Im}(\mathbb{P}) \,\big\}.$$

The kill set measures the kernel of the linear map $a \mapsto \mathcal{C}_{a,*}(p) \in \text{Hom}(\text{Im}(\mathbb{P}), \text{End}(K_p))$. At sedenion the kill set is the canonical $\mathfrak{q}_p = \iota_1(V_p) \oplus \mathbb{R}\ell \oplus \iota_2(V_p)$ of dimension 5, where $V_p = \text{span}_{\mathbb{R}}\{x, y\} \subset \mathbb{S}$ [2, R133]. The pathion lift of this statement is the central object of this paper, and it splits into two distinct theorems with different rigor sources.

Status conventions. Proof vocabulary (prove, proved, proven) is reserved in this paper for claims established symbolically. Computational witnesses carry a separate vocabulary: a certification states its epistemic basis, either a numerical tolerance or *exact* (exhaustive finite verification or exact arithmetic), and is never restated as a proof. Where a theorem's two directions have different rigor sources, each direction's basis is stated in place; in particular, the necessity directions of Theorems 4.1 and 4.3 are numerical certifications, not proofs. Citations of the form "v47-A22" or "framework state" refer to the author's internal research ledger of promoted results; the sedenion-level foundations are published in the two companion papers [1, 2], and framework-ledger material cited here should be read as announced, with proofs of record in that ledger.

3 Pathion spectrum stratification

The defining feature distinguishing pathion from sedenion is the loss of orbit invariance of the Gram spectrum. At sedenion, $\mathrm{Spec}(G_p)$ is the universal triple $\{0^4, 2^8, 4^4\}$ for every aphanic p. At pathion this universality fails:

Proposition 3.1 (Spectrum Non-Invariance)**.** *There exist aphanic $p, p' \in \mathcal{P}_{\mathbb{P}}^{\mathrm{aph}}$ with $\dim K_p = \dim K_{p'} = 4$ and $\|p\| = \|p'\|$ such that $\mathrm{Spec}(G_p) \neq \mathrm{Spec}(G_{p'})$. Specifically, dyadic basepoints of the form $p = e_i + e_j$ (in a Stratum II configuration of Proposition 2.2; canonical representative $p = e_1 + e_{26}$) have the three-valued spectrum $\{0^4, \|p\|^2\, 24, 2\|p\|^2\, 4\}$, while random aphanic basepoints have an eight-valued spectrum $\{0^4, \lambda_1^4, \ldots, \lambda_7^4\}$.*

The structural consequence is computational: at dyadic basepoints, the pseudoinverse G_p^+ admits a closed-form Lagrange-3 polynomial representation $G_p^+ = \frac{7}{4\|p\|^4} G_p - \frac{3}{4\|p\|^6} G_p^2$, while at generic aphanic basepoints the pseudoinverse must be obtained from an eigendecomposition. This distinction is the technical reason why the kill set has different dimensions in the two regimes (Theorems 4.1 and 4.3 below).

Orbit-level refinement of Proposition 3.1. The non-invariance recorded above has since been resolved into a partial orbit classification in subsequent framework work, which we record here as cross-reference rather than reprove. Let $\mathcal{A}, \mathcal{B} \subset \mathcal{P}_{\mathbb{P}}^{\mathrm{aph}}$ denote, respectively, the loci of pathion-aphanic basepoints with $\|p\|^2 = 2$ and Gram spectrum $\{0^{12}, 2^8, 4^{12}\}$ (*Class A*) or $\{0^4, 2^{24}, 4^4\}$ (*Class B Stratum II*). Two facts are established outside the present paper:

1. **Class A complement spectrum theorem (R138).** On the six named dyadic Class A subcases enumerated in the framework state, $\mathrm{Spec}(G_p) = \{0^{12}, 2^8, 4^{12}\}$ uniformly; this spectrum is the defining Class A property. What R138 proves is the *complement* spectrum theorem: for the complement operators $G_k := M_k^\top M_k$ (with $M_k = A - T_k$, $k \in \{2, 4, 6\}$ indexing the three outer cosets), $\mathrm{Spec}(G_k) = \{0^{12}, 2^8, 4^{12}\}$ as well, via the per-octonion-sub identity $G_k|_{\mathbb{O}_p} = 2I_8$ combined with $G_k|_{\mathbb{O}_p^\perp} = 4\,\pi_p^{\mathrm{gr}}$, where π_p^{gr} is the rank-4 orthogonal projector onto the graph of the isometry $+\Phi_p$ within each outer coset (not the kernel projector π_p of this paper). Proven and promoted in the framework state at v46-C18 close.

2. **Class B Stratum II three-orbit classification (R140).** Every $p \in \mathcal{B}$ lies in exactly one of three G_2-orbits: *canonical* $G_2 \cdot (e_1 + e_{26})$, *mirror* $G_2 \cdot (e_{11} + e_{20})$, or *broken-mirror balanced* $G_2 \cdot p_3$, where $p_3 = \frac{1}{\sqrt{2}}(e_1 + e_{25} + e_{10} + e_{18})$ satisfies $\mathrm{Spec}(G_{p_3}) = \{0^4, 2^{24}, 4^4\}$ together with $\mathrm{cb}^2_{\mathrm{can}}(p_3) = 12$ and $\mathrm{cb}^2_{\mathrm{mir}}(p_3) = 4$. The third orbit was discovered as a refutation of an earlier two-layout exhaustion conjecture and is not in either of the two named layouts of R139 (the prior two-layout sufficient-condition theorem at v47-A7). R140 strictly extends R139 without modifying it; R139's two-layout scope remains correct as

a sufficient condition. Proven and promoted in the framework state at v47-A22 close via two new lemmas: a G_2-equivariance lift of the Class B variety and a sign-correlation rule $\mathrm{sign}(\alpha_1)\mathrm{sign}(\beta_2) = \mathrm{sign}(\beta_1)\mathrm{sign}(\alpha_2)$ derived from the collapse of the three independent Class B trace conditions to a single parity equation $P = 1$, where $P := s_1 s_{25} s_{10} s_{18}$.

The eight-valued generic-aphanic class (*Class C*) remains open as an orbit *classification*; its separability by a natural 10-tuple of continuous G_2-invariants is, however, resolved *negatively* on a local/generic neighborhood by R142, and the residual rank-3 deficiency is then closed *positively* by an explicit 13-tuple of local coordinates on a dense open set (R143). Both are recorded after the R141 paragraph below. The $\rho = 3$ single-component-redundancy classes (*D*, *E*, *F*) are closed by R141 below.

Structural anatomy of case (3) (v47-A24 to v47-A29). The closed-form anatomy of the third orbit on its broken-mirror balanced two-parameter family

$$p(a,b) \;:=\; a(e_1 + e_{25}) + b(e_{10} + e_{18}) \in \mathbb{P}, \qquad (a,b) \in \mathbb{R}^2,$$

was developed in subsequent framework work (v47-A24 through v47-A29). For each of the three non-trivial $16+16$ partitions $U_+^* \oplus U_-^*$ of $\mathbb{P}$, namely the *canonical* ($Q_{00} \oplus Q_{11}$, $Q_{01} \oplus Q_{10}$), the *mirror* ($Q_{00} \oplus Q_{10}$, $Q_{01} \oplus Q_{11}$), and the *sed-half* ($Q_{00} \oplus Q_{01}$, $Q_{10} \oplus Q_{11}$), the squared off-block commutator-balance functional $\mathrm{cb}_*^2(p) := \|G_p[U_+^*, U_-^*]\|_F^2$ admits a closed form on the family:

$$\begin{aligned}
\mathrm{cb}^2_{\mathrm{can}}(a,b) &= 48\,a^2b^2,\\
\mathrm{cb}^2_{\mathrm{mir}}(a,b) &= 16\,(3a^4 - 5a^2b^2 + 3b^4),\\
\mathrm{cb}^2_{\mathrm{sed}}(a,b) &= 16\,(3a^4 - 2a^2b^2 + 3b^4).
\end{aligned}$$

At the balanced point p_3 (i.e. $a = b = 1/\sqrt{2}$) these specialize to 12, 4, and 16 respectively, recovering the numerical values cited above. The canonical formula is derived from path–Cayley–Dickson multiplication identities via a structural ab-purity argument on the column-pair signature of L_p and an octonion associator-into-conjugation cancellation $(I + K_{\mathbb{O}}) \circ [L_{e_1}, R_{e_2}]$ on the within-sed-half edges (framework state at v47-A25 close). The double-swap diagonal-block edges

$$x_{Q_{00},Q_{11}}(a,b) \;=\; x_{Q_{01},Q_{10}}(a,b) \;=\; 8(3a^4 - 5a^2b^2 + 3b^4)$$

underlying $\mathrm{cb}^2_{\mathrm{mir}}$ and $\mathrm{cb}^2_{\mathrm{sed}}$ admit the operator-polynomial form

$$G_p[Q_{00}, Q_{11}] \;=\; a^2 \cdot A \;+\; b^2 \cdot B, \qquad A := I + L_{e_1} R_{e_1}, \qquad B := K_{\mathbb{O}}\,(I + L_{e_2} R_{e_2}),$$

where $A = 2P_{\{1,e_1\}^\perp}$ and $B = -2P_{\{1,e_2\}^\perp}$ on $\mathbb{O}$ (twice and negative-twice the orthogonal projectors onto the 6-dimensional subspaces orthogonal to the indicated pairs), with $\|A\|_F^2 = \|B\|_F^2 = 24$ and $\langle A, B\rangle_F = -20$, the inner product traced to the 5 shared oct-basis directions $\{e_3, e_4, e_5, e_6, e_7\}$ in $\{e_1\}^\perp \cap \{e_2\}^\perp \cap \mathrm{Im}(\mathbb{O})$ (framework state at v47-A28 close). The mirror and sed-half quartics follow as partition assemblies of the six edge weights with each unordered quad pair appearing in exactly two of the three partitions (framework state at v47-A29 close).

At p_3 specifically, the canonical 16×16 off-block $G_{p_3}[U_+^{\mathrm{can}}, U_-^{\mathrm{can}}]$ has rank 6, singular spectrum $\sigma = (2,2,1,1,1,1)$, and Frobenius norm squared $\|\cdot\|_F^2 = 12$, with the closed operator-skeleton identity on the family

$$G_p[U_+^{\mathrm{can}}, U_-^{\mathrm{can}}]^{(\mathrm{reord})} \;=\; ab \cdot (M \oplus N^\top)$$

after a natural column reorder $P_{\mathrm{col}} \in O(16)$ of U_-^{can} from (Q_{01}, Q_{10}) to (Q_{10}, Q_{01}). Here

$$\begin{aligned}
M &:= L_{e_1} R_{e_2} K_{\mathbb{O}} + K_{\mathbb{O}} L_{e_2} L_{e_1} K_{\mathbb{O}} - K_{\mathbb{O}} L_{e_2} R_{e_1} - L_{e_1} L_{e_2}\\
N &:= K_{\mathbb{O}} R_{e_2} L_{e_1} + R_{e_1} R_{e_2} - K_{\mathbb{O}} R_{e_1} L_{e_2} - R_{e_2} R_{e_1}
\end{aligned}$$

have rank 2 and rank 4 on the quaternion subalgebra $\mathbb{H} = \mathrm{span}(1, e_1, e_2, e_3) \subset \mathbb{O}$, with M acting as a single ϵ-tensor on the pair (e_1, e_2) and N acting as a direct sum of two ϵ-tensors on the pairs $(1, e_3)$ and (e_1, e_2). Both M and N vanish on $\mathbb{H}^\perp \subset \mathbb{O}$ (framework state at v47-A26 close).

The canonical and mirror partitions are not related by any rank-preserving σ-bridge: at every basepoint $p \in G_2 \cdot p_3$ in the third orbit, $\mathrm{rank}\big(G_p[U_+^{\mathrm{can}}, U_-^{\mathrm{can}}]\big) = 6$ while $\mathrm{rank}\big(G_p[U_+^{\mathrm{mir}}, U_-^{\mathrm{mir}}]\big) = 4$; since rank is invariant under orthogonal $R, C \in O(16)$, signed row/column permutations, scalar rescaling, and pathion automorphisms $\phi \in \mathrm{Aut}(\mathbb{P})$, no identity of the form $G_p[U_+^{\mathrm{mir}}, U_-^{\mathrm{mir}}] = \varepsilon \cdot R \cdot G_p[U_+^{\mathrm{can}}, U_-^{\mathrm{can}}] \cdot C$ exists. The four magnitude-1 singular values are nevertheless common to the canonical and mirror off-blocks at p_3; the canonical off-block carries two additional magnitude-2 singular values originating from N's $(1, e_3)$ ϵ-tensor block (the unique-to-β real-to-e_3 coupling), which has no mirror-partition counterpart (framework state at v47-A27 close).

The Class A complement spectrum theorem R138 and the Class B Stratum II three-orbit classification theorem R140 recorded above are unchanged by this anatomy. No new R-promotion is claimed at v47-A24 through v47-A29; the closed forms and operator identities above supply structural derivations of the third-orbit anatomy that had previously been available in framework state via symbolic computation. The eight-valued class C remains open as a classification (its 10-tuple local/generic non-separation is recorded as R142, after the R141 paragraph); classes D, E, F are resolved by R141 in the following paragraph.

Class D/E/F cross-quad trace fingerprint (R141). The preceding three-valued spectrum classes (A, B) do not exhaust the pathion-aphanic locus. Let the oct-rank $\rho(p) := \mathrm{rank}_{\mathbb{O}}(\alpha_1, \beta_1, \alpha_2, \beta_2)$ denote the number of linearly independent octonion-imaginary components in the four-quad decomposition of p. Classes A ($\rho = 1$) and B ($\rho = 2$) have been treated above. The remaining cases with $\rho = 3$ (i.e., exactly one single-component redundancy $p_{\mathrm{dep}} = \pm p_{\mathrm{tgt}}$ among the four quad directions) produce *quadratic-irrational* Gram spectra (eigenvalues involving $\sqrt{2}$, $\sqrt{3}$, or $\sqrt{5}$) and split into three classes according to the position of the redundancy:

- **Class D** (ABcross = 1, SedCross = 1): redundancy crosses both the (α, β) pair index and the sedenion-half index, e.g. $\beta_2 = \alpha_1$.
- **Class E** (ABcross = 1, SedCross = 0): redundancy crosses the pair index within one sedenion half, e.g. $\beta_1 = \alpha_1$.
- **Class F** (ABcross = 0, SedCross = 1): redundancy is a proportionality within one pair across sedenion halves, e.g. $\alpha_2 \parallel \alpha_1$ after aphanic projection.

We record the following cross-referenced theorem, established outside the present paper at framework state v47-B10 close.

3. **Class D/E/F cross-quad trace fingerprint, canonical golden-ratio block, and full-versus-local eigenspace split (R141).** Let $p \in \mathcal{P}_{\mathbb{P}}^{\mathrm{aph}}$ at $\|p\|^2 = 2$ with $\rho(p) = 3$ and class label determined by the $(\mathrm{ABcross}, \mathrm{SedCross})$ predicate above. Recall the oct-quad partition $Q_{00}, Q_{01}, Q_{10}, Q_{11}$ of $\mathbb{P}$ and the canonical partition $U_+^{\mathrm{can}} = Q_{00} \oplus Q_{11}$, $U_-^{\mathrm{can}} = Q_{01} \oplus Q_{10}$ already used above.

   (i) Universal block form. For any $\rho = 3$ class,

$$G_p\big|_{U_+^{\mathrm{can}}} = \begin{bmatrix} 2I_8 & M_p \\ M_p^\top & 2I_8 \end{bmatrix}, \qquad M_p := G_p[Q_{00}, Q_{11}],$$

   where the diagonal $2I_8$ blocks follow from the Per-Quad-Diagonal-Block Lemma (A12-2.1) at aphanic + $\|p\|^2 = 2$.

   (ii) Trace fingerprint (G_2-orbit invariant). $\mathrm{tr}(M_p) = 6$ for Class D and $\mathrm{tr}(M_p) = 0$ for Classes E and F. The Frobenius norm satisfies $\|M_p\|_F^2 = 8$ at the canonical basepoints of

all three classes (hence on their G_2-orbits, by the equivariance below). Both invariants are preserved under the lifted G_2-action $g_P = \mathrm{diag}(g_O \times 4) \in \mathrm{Aut}(\mathbb{P})$ for $g_O \in G_2 = \mathrm{Aut}(\mathbb{O})$, via the orthogonal similarity $M_{g_P \cdot p} = g_O \, M_p \, g_O^\top$ (Lemma R141.1).

(iii) Class D two-orbit ς-stratification. Across the 210 oct-permutation Class D variants $\{(a,b,c) : a,b,c \in \{1,\dots,7\},\ \text{distinct}\}$, the singular-value tuple $\varsigma(M)$ takes exactly two values, each a single G_2-orbit:

- *Fano-triple orbit* ($42 = 7$ Fano lines $\times$ 3! members): $\varsigma(M) = (0,\, 1/\varphi,\, 1^5,\, \varphi)$ with $\varphi = (1+\sqrt{5})/2$.
- *Non-Fano orbit* ($168 = 210 - 42$ members): $\varsigma(M) = (0^2,\, 1/\varphi,\, 1^3,\, \sqrt{2},\, \varphi)$.

The Fano orbit is characterized by the condition that the redundancy triple $(\beta_1, \alpha_2, \alpha_1)$ lies on a Fano line of the octonion multiplication table.

(iv) Canonical golden-ratio block. At the canonical Class D Fano-orbit representative $p_D^{\mathrm{can}} = (1/\sqrt{2})(e_1 + e_{11} + e_{18} + e_{25})$, the 8×8 block M decomposes as

$$M = E_{10}^{(2)} \oplus \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \oplus I_4,$$

where $E_{10}^{(2)} = \left[\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right]$, and the golden-ratio singular values originate from the 2×2 sub-block $M_2 := \left[\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right]$ with $M_2^\top M_2$ having eigenvalues $(3 \pm \sqrt{5})/2 = \varphi^2,\, 1/\varphi^2$.

(v) Full-versus-local V_1 split. At canonical Class D, the full G_p eigenvalue-1 eigenspace $V_1 \subset \mathbb{P}$ (dimension 8) splits perfectly: $\dim(V_1 \cap U_+^{\mathrm{can}}) = \dim(V_1 \cap U_-^{\mathrm{can}}) = 4$. The H-localized eigenvalue-1 multiplicity of 5 on U_+^{can} decomposes as 4 genuine (from $V_1 \cap U_+^{\mathrm{can}}$) + 1 spurious (restriction artifact not an eigenvector of the full G_p).

(vi) Class E/F trace-zero invariance. For canonical Class E: $\mathrm{tr}(M_p) = 0$ by a cyclic-trace cancellation in the per-K-sum expansion, where opposing K-positions generate $\pm L_{e_1} R_{e_1} K$ terms with inverted sign. For canonical Class F: $\mathrm{tr}(M_p) = 0$ by three independent cancellation mechanisms (identity, cyclic K-flip, and non-fixing residual trace-zero permutations) applied to the per-K-sum with composite $\beta_2 = \sqrt{2/3}\, e_3 - (1/\sqrt{6})\, e_1$. Both invariances lift to the full G_2-orbit by Lemma R141.1 (Lemma R141.3).

The proof stack consolidates v47-B6 (cross-quad identification), v47-B7a (Fano-line signed-permutation trace algebra), v47-B7b (bit-exact block form and golden-ratio discovery), v47-B7c (V_1 split and Class F per-K closure), and v47-B9 (formal G_2-equivariance, finite-exhaustive ς-stratification, and E/F corollary). Promoted and recorded in the framework state at v47-B10 close. Class D partitions into exactly two G_2-orbits; a continuous ς-classification of the full $\rho = 3$ locus is *not* claimed. A representation-theoretic explanation of why Fano-triple $\Longrightarrow$ golden-ratio singular values is an open question.

Class C 10-tuple local/generic incompleteness (R142). The generic eight-valued class C ($\rho = 4$: the four quad directions $\alpha_1, \beta_1, \alpha_2, \beta_2$ are octonion-linearly independent, $\dim K_p = 4$, and $\mathrm{Spec}(G_p)$ takes eight distinct values) is, unlike the $\rho \le 3$ classes above, not resolved into a finite orbit list here. What *is* established (and recorded as a cross-reference to framework state v47-C8) is a sharp obstruction to separating its G_2-orbits by a natural tuple of continuous invariants. Let

$$F = \big(I_2,\ I_4,\ \mathrm{tr}\, G_p^2,\ \mathrm{tr}\, G_p^3,\ \mathrm{tr}\, G_p^4,\ \mathrm{tr}\, G_p^5,\ \mathrm{tr}\, G_p^6,\ J_3,\ J_4,\ J_5\big) : M \to \mathbb{R}^{10}$$

be the 10-tuple of continuous G_2-invariant functions on the open Class C locus M, with $I_2 = \|\alpha_1\|^2 + \|\beta_2\|^2$, $I_4 = \langle \alpha_1, \beta_2 \rangle + \langle \beta_1, \alpha_2 \rangle$, $J_3 = \langle [\alpha_1, \beta_1], [\alpha_2, \beta_2] \rangle$, $J_4 = \langle [\alpha_1, \beta_1, \alpha_2], \beta_2 \rangle$, $J_5 = \langle \alpha_1, \alpha_2 \rangle \langle \beta_1, \beta_2 \rangle - \langle \alpha_1, \beta_1 \rangle \langle \alpha_2, \beta_2 \rangle$, and $G_2 = \mathrm{Aut}(\mathbb{O})$ acting by $g_P = \mathrm{diag}(g_O \times 4)$.

4. **Class C 10-tuple local/generic incompleteness (R142).** On a nonempty open neighborhood of an exhibited *regular* Class C point, the local manifold M has dimension 27, the G_2-orbit has dimension $d_O = 14$, and $\operatorname{rank} dF|_{TM} = 10$. Consequently the local G_2-quotient has dimension $27 - 14 = 13 > 10$, and F *does not separate the G_2-orbits of* M: there exist distinct G_2-orbits with equal F-value, and the local/generic separation deficiency is exactly $13 - 10 = 3$.

Proof sketch (exact single-point witnesses; framework v47-C7/C7b/C8). The three dimension counts are certified at one *exact rational* Class C witness q (an integer aphanic pathion with $\langle s_1, s_2\rangle = 0$, $\|s_1\|^2 = \|s_2\|^2$, oct-rank 4, $\dim\ker L_q = 4$, and eight distinct G_q-eigenvalues), each rank established by a nonzero *exact* minor rather than a floating-point threshold: **(L1)** the 3×3 constraint-gradient minor has determinant $80 \neq 0$, so $\dim M = 27$; **(L2)** the 21×30 matrix of the $\mathfrak{g}_2$ derivations $D_{a,b} = [L_a, L_b] + [L_a, R_b] + [R_a, R_b]$ applied block-diagonally has a nonzero 14×14 minor, and since $d_O \leq \dim \mathfrak{g}_2 = 14$ this forces $d_O = 14$ (so q is regular, with finite isotropy); **(L3)** the tangent-projected Jacobian $dF|_{TM}$ has a nonzero 10×10 minor, so $\operatorname{rank} dF|_{TM} = 10$ (the Jacobian was independently re-derived along a truncated-polynomial path, 300/300 entries agreeing, with an integer 13×13 cross-check of $[\,C;\, dF\,]$). Two further exact certificates close the descent: **(K1)** $dF \cdot \mathrm{Orb}^\top = 0$, the single-point confirmation of the (independently established) global G_2-invariance of F, so the 14-dimensional orbit lies in $\ker dF$; and **(K2)** $C \cdot \mathrm{Orb}^\top = 0$, so the orbit is tangent to M. Each construction is homogeneous in p, so the ranks at the integer point q equal those at the unit-norm point $p_0 = q/\sqrt{c} \in M$ (scaling multiplies each matrix row by a nonzero constant and fixes $T = \ker C$). By the constant-rank theorem the fiber of F is locally a $27 - 10 = 17$-manifold containing the 14-dimensional orbit, hence a 3-parameter family of distinct orbits sharing F; equivalently, the 13-dimensional slice (slice theorem, G_2 compact) cannot inject continuously into $\mathbb{R}^{10}$ (invariance of domain). No floating-point quantity is load-bearing. Promoted and recorded in the framework state at v47-C8 close. □

Scope. The statement is *local/generic* (the dimension counts are certified on the maximal-rank neighborhood of one regular witness, not as a density claim) while the non-separation conclusion itself is unconditional. It is confined to *pathion Class C*. It does *not* classify the Class C G_2-orbits, and does *not* assert that F is uninformative: the 10-tuple is functionally independent (its Jacobian has full rank 10) and accounts for 10 of the 13 local orbit coordinates. (An *explicit* finite-distance pair of distinct orbits with equal F is exhibited in the constructive remark below; a *continuous, non-inversion* such family is left open.) The global "deficiency 3 at all basepoints" is, separately, an empirical framework observation, not part of R142.

Constructive collision witness (exact). The local/generic obstruction of R142 is complemented by an *exact, finite-distance* collision. Let q be the exact rational Class C witness used in the R142 certification. Its central inverse $-q$ is again Class C, and $F(q) = F(-q)$ *exactly*, because every component of the 10-tuple is an *even* function of p: I_2, I_4 are quadratic, $\operatorname{tr} G_p^k$ is even since $G_{-p} = L_{-p}^\top L_{-p} = L_p^\top L_p = G_p$, and J_3, J_4, J_5 have degree 4. (This evenness holds for every pathion; Class C membership of both points is an additional satisfied constraint.) The two representatives are nevertheless *not* G_2-equivalent: the associative three-form $\varphi_1(p) = \langle[\alpha_1, \beta_1], \alpha_2\rangle$ is G_2-invariant (as $\mathrm{Aut}(\mathbb{O})$ preserves the octonion commutator and the inner product) and *odd*, and at the witness $\varphi_1(q) = -14$ while $\varphi_1(-q) = +14$; were $-q = g \cdot q$ for some $g \in G_2$ we would have $\varphi_1(-q) = \varphi_1(q)$, a contradiction. Hence equality of the full 10-tuple does not imply G_2-orbit equivalence. The remaining invariants agree, e.g. $\operatorname{tr} G_q^2 = 234145504$, $J_3 = 98560$, $J_4 = 48128$, $J_5 = -70217$ (all equal for q and $-q$), and the pair is at finite distance $\|q - (-q)\|^2 = 8120 = 4\|q\|^2$. All values are exact integers; no floating point enters. This realizes the non-separation by an exact *parity / orientation* blindness (F is even while the orbit-separating associative three-forms are odd) and is confined to the *central-inversion*

collision; a continuous, non-inversion finite-distance family and a global orbit classification both remain open.

Class C 13-tuple local full-rank (R143). The local separation deficiency of exactly 3 counted by R142 is closed, generically, by the three associative G_2-orientation three-forms. Adjoin to the 10-tuple F the forms $\varphi_1(p) = \langle[\alpha_1, \beta_1], \alpha_2\rangle$ (the odd invariant of the collision witness above), $\varphi_2(p) = \langle[\alpha_1, \beta_1], \beta_2\rangle$, and $\varphi_3(p) = \langle[\alpha_1, \alpha_2], \beta_2\rangle$ (all G_2-invariant because $\mathrm{Aut}(\mathbb{O})$ preserves the octonion commutator and the inner product) and set

$$F_{13} = \big(I_2,\, I_4,\, \mathrm{tr}\, G_p^2,\, \ldots,\, \mathrm{tr}\, G_p^6,\, J_3,\, J_4,\, J_5,\, \varphi_1,\, \varphi_2,\, \varphi_3\big) : M \to \mathbb{R}^{13}.$$

5. **Class C 13-tuple local full-rank (R143).** On a *dense* Zariski-open subset of the Class C stratum, $\mathrm{rank}\, dF_{13}|_{TM} = 13$; equivalently F_{13} is a full-rank local coordinate system for the 13-dimensional local Class C quotient there, with fibers the G_2-orbits. The three forms $\varphi_1, \varphi_2, \varphi_3$ supply exactly the 3 orbit directions invisible to the R142 10-tuple.

Proof sketch (exact; no symbolic determinant; framework v47-C12 / C13). The augmented Jacobian $A = [\,C;\, dF_{13}\,] \in (\mathbb{Z}[p])^{16\times 30}$ has integer-polynomial entries (constraint gradients linear; spectral rows $k\,\mathrm{tr}(G_p^{k-1}(B_m^\top L_p + L_p^\top B_m))$; the octonionic and φ rows gradients of degree-≤ 4 integer-coefficient invariants). A distinguished 16×16 minor $M_J \in \mathbb{Z}[p]$ evaluates to $-1.9042\ldots \times 10^{43} \neq 0$ at the exact rational Class C witness q, so $M_J \not\equiv 0$; hence $U = \{p : M_J(p) \neq 0\}$ is a nonempty Zariski-open subset on which $\mathrm{rank}\, A = 16$ (corroborated by nonzero 16×16 minors at the witness, $\|p\|^2 = 70$, and at two further independent Class C points, $\|p\|^2 = 78, 106$). By $\mathrm{rank}(D|_{\ker C}) = \mathrm{rank}[\,C;\, D\,] - \mathrm{rank}\, C$, the full row rank forces $\mathrm{rank}\, C = 3$ and $\mathrm{rank}\, dF_{13}|_{TM} = 13$ on U. All 13 components are G_2-invariant, so the 14-dimensional orbit (the generic value $d_O = 14$ from R142) lies in $\ker dF_{13}|_{TM}$; with rank 13 the kernel is exactly the orbit, so F_{13} is a local submersion onto $\mathbb{R}^{13}$ with orbit fibers: local coordinates / local orbit separation by the implicit function theorem.

Density (Lemma L-C13.1, framework v47-C13). The Class C stratum is cut out by $\langle s_1, s_2\rangle = 0$, $\|s_1\|^2 = \|s_2\|^2$, $\|p\|^2 = 2$, that is, it is the Stiefel manifold $V_2(\mathbb{R}^{15})$ of orthonormal 2-frames in $\mathbb{R}^{15}$. By Witt's extension theorem $SO(15)$ acts transitively on such frames, so $V_2(\mathbb{R}^{15})$ is a single orbit of the connected group $SO(15,\mathbb{R})$ and its complexification $V_2(\mathbb{C}^{15})$ is a single $SO(15,\mathbb{C})$-orbit, an irreducible affine variety of dimension $2\cdot 15 - 3 = 27$, in which $V_2(\mathbb{R}^{15})$ is Zariski-dense. A nonzero polynomial on an irreducible variety has nowhere-dense zero set, so U is in fact a *dense* Zariski-open subset, and the full-rank locus is dense in the Class C stratum. No floating-point quantity is load-bearing. Promoted at v47-C12; scope sharpened from "nonempty" to "dense" Zariski-open at v47-C13. □

Scope. R143 is the positive companion to R142: where R142 *counts* the local separation deficiency ($13-10=3$), R143 *exhibits* the three explicit invariants φ_i that remove it and furnish local coordinates on a dense open set. The statement is *local* (separation of nearby orbits via the implicit function theorem) and does *not* assert *global* orbit separation: the central inversion $q \mapsto -q$ shows the odd φ_i separate that particular pair, but global injectivity on $V_2(\mathbb{R}^{15})/G_2$ and a continuous, non-inversion collision both remain open. Confined to pathion Class C; the rank may drop on the nowhere-dense closed set $\{M_J = 0\}$.

4 Two pathion Q-centralizer vanishing theorems

4.1 The universal kill set

Theorem 4.1 (Pathion Universal Kill Set)**.** *Let $p \in \mathcal{P}_{\mathbb{P}}^{\mathrm{aph}}$ with $\dim K_p = 4$. Then*

$$\mathcal{K}(p) \;=\; \mathfrak{q}_p^{\mathrm{univ}} \;:=\; \mathrm{span}_{\mathbb{R}}\{p,\, \ell_{16},\, \ell_{16}\cdot p\}, \qquad \dim_{\mathbb{R}} \mathfrak{q}_p^{\mathrm{univ}} = 3.$$

(*Sufficiency is proved symbolically; necessity is a numerical certification, Appendix A.*)

The proof proceeds via a centralization chain. Define

$$G_p = L_p^\top L_p, \qquad \pi_p = I - G_p G_p^+.$$

Lemma 4.2 (Centralization Chain)**.** *For any $a \in \mathrm{Im}(\mathbb{P})$, the following implications hold:*

$$\{L_a, L_p\} = 0 \implies [L_a, L_p^2] = 0 \iff [L_a, G_p] = 0 \implies [L_a, \pi_p] = 0 \implies \mathcal{C}_{a,b}(p) = 0 \; \forall b.$$

Proof. The first implication is two lines: $\{L_a, L_p\} = 0$ gives $L_a L_p = -L_p L_a$, hence $L_a L_p^2 = (L_a L_p) L_p = -(L_p L_a) L_p = -L_p (L_a L_p) = L_p^2 L_a$.

The second equivalence uses $G_p = -L_p^2$, which follows from $L_p^\top = -L_p$ for $p \in \mathrm{Im}(\mathbb{P})$ (since $\bar{p} = -p$ implies $L_p^\top = L_{\bar{p}} = -L_p$).

The third implication uses the spectral theorem: π_p is the orthogonal projector onto $\ker G_p$ and is therefore a polynomial in G_p over the spectrum of G_p. Hence anything commuting with G_p commutes with π_p.

The final implication is the formal collapse of (2): using $[L_a, \pi_p] = 0$ to move π_p across L_a, and $G_p^+ \pi_p = \pi_p G_p^+ = 0$ (since $G_p^+ \in \mathbb{R}[G_p]$ vanishes on $\ker G_p$),

$$\begin{aligned}
\mathcal{C}_{a,b}(p) &= \pi_p (L_b G_p^+ L_a - L_a G_p^+ L_b) \pi_p \\
&= \pi_p L_b G_p^+ (L_a \pi_p) - (\pi_p L_a) G_p^+ L_b \pi_p \\
&= \pi_p L_b (G_p^+ \pi_p) L_a - L_a (\pi_p G_p^+) L_b \pi_p \\
&= 0.
\end{aligned}$$

□

Proof of Theorem 4.1. **Sufficiency** ($\mathfrak{q}_p^{\mathrm{univ}} \subseteq \mathcal{K}(p)$): By Lemma 4.2 it suffices to show $\{L_a, L_p\} = 0$ for $a \in \{p, \ell_{16}, \ell_{16} \cdot p\}$. For $a = p$, $\{L_p, L_p\} = 2L_p^2 = -2L_p^\top L_p = -2G_p$, so $[L_p, L_p^2] = 0$ trivially and the conclusion follows directly from the rest of Lemma 4.2. For $a = \ell_{16}$ and $a = \ell_{16} \cdot p$, the anti-commutation is the content of the Pathion Clifford Triad Theorem (Theorem 5.1), proven in Section 5.

Necessity ($\mathcal{K}(p) \subseteq \mathfrak{q}_p^{\mathrm{univ}}$): Construct the symmetric matrix $\Gamma_{\mathcal{C}} \in \mathbb{R}^{31 \times 31}$ with entries $(\Gamma_{\mathcal{C}})_{a,a'} = \sum_b \langle \mathcal{C}_{a,b}(p), \mathcal{C}_{a',b}(p) \rangle_F$ where the sum runs over a basis of $\mathrm{Im}(\mathbb{P})$ and the inner product is the Frobenius product on $\mathrm{End}(K_p)$. The kill set is exactly $\ker \Gamma_{\mathcal{C}}$. Computational certification across $N = 16$ random aphanic basepoints (Appendix A) shows $\dim \ker \Gamma_{\mathcal{C}} = 3$ and that $\mathfrak{q}_p^{\mathrm{univ}} \subseteq \ker \Gamma_{\mathcal{C}}$ at machine precision. The nullity is an integer conclusion read from a floating-point computation: this direction is a certification, not a proof. Granting the certified nullity, equality follows by dimension count. □ □

4.2 The dyadic extension

Theorem 4.3 (Pathion Q-Centralizer Vanishing at Dyadic Basepoint)**.** *Let $p \in \mathcal{P}_{\mathbb{P}}^{\mathrm{aph}}$ with $\mathrm{Spec}(G_p) = \{0^4, \|p\|^2 {}^{24}, 2\|p\|^2 {}^{4}\}$. Then*

$$\mathcal{K}(p) \;=\; \mathfrak{q}_p^{(32)} \;:=\; \iota_1(V_p) \oplus \mathbb{R}\ell_{16} \oplus \iota_2(V_p), \qquad \dim_{\mathbb{R}} \mathfrak{q}_p^{(32)} = 5,$$

where $V_p = \mathrm{span}_{\mathbb{R}}\{s_1, s_2\}$ for $p = \iota_1(s_1) + \iota_2(s_2)$. (Sufficiency is proved symbolically; necessity is a numerical certification at canonical Stratum-II, Appendix A.)

Proof. **Sufficiency** ($\mathfrak{q}_p^{(32)} \subseteq \mathcal{K}(p)$): For $a \in \mathfrak{q}_p^{\mathrm{univ}}$, $[L_a, G_p] = 0$ follows from the Pathion Clifford Triad Theorem 5.1 via the centralization chain Lemma 4.2. For $a \in \mathfrak{q}_p^{\mathrm{anti}}$, $[L_a, G_p] = 0$ holds at every dyadic basepoint by the Two-Regime Dyadic Closure Theorem 7.17 (proven in §7). By linearity in a and the decomposition $\mathfrak{q}_p^{(32)} = \mathfrak{q}_p^{\mathrm{univ}} \oplus \mathfrak{q}_p^{\mathrm{anti}}$, $[L_a, G_p] = 0$ holds for every $a \in \mathfrak{q}_p^{(32)}$, hence by Lemma 4.2, $[L_a, \pi_p] = 0$ and $\mathcal{C}_{a,b}(p) = 0$ for all $b \in \mathrm{Im}(\mathbb{P})$. Equivalently, $\mathfrak{q}_p^{(32)} \subseteq \mathcal{K}(p)$. See also Corollary 7.18.

Necessity ($\mathcal{K}(p) \subseteq \mathfrak{q}_p^{(32)}$): By the same nullity argument as in Theorem 4.1, using a $\Gamma_{\mathcal{C}}$-spectrum analysis whose details are recorded in the certification appendix. Computational certification across canonical Stratum-II dyadic basepoints (Appendix A) shows $\dim\ker\Gamma_{\mathcal{C}} = 5$ and $\mathfrak{q}_p^{(32)} \subseteq \ker\Gamma_{\mathcal{C}}$ at machine precision. As in Theorem 4.1, this direction is a certification, not a proof; granting the certified nullity, equality follows by dimension count. □ □

Remark 4.4. The dim-5 extension at dyadic basepoints does *not* arise from additional Clifford anti-commutation with the universal triad: as we show in Section 6, the anti-symmetric extensions form a Clifford pair internally (Theorem 6.3 (2)) but do not anti-commute with the universal triad. The mechanism for their centralization with G_p at dyadic basepoints is the two-regime closure of §7: at canonical Stratum-II, L_w is block-diagonal in the G_p-eigenspace decomposition (Theorem 7.13); at the CD-aligned subfamily, $\{L_w, L_p\} = 0$ holds directly (Lemma 7.14); the u-side closure $\{L_u, L_p\} = 0$ holds across the entire dyadic family (Lemma 7.6). The closed-form Lagrange-3 pseudoinverse $G_p^+ = \frac{7}{4\|p\|^4}G_p - \frac{3}{4\|p\|^6}G_p^2$ is consistent with the closure but not load-bearing: every step of the two-regime derivation works directly from the half-localization, block-diagonality, or octonion-flexibility identities without invoking the polynomial form of G_p^+. At generic aphanic, all three regime mechanisms fail (Remarks 7.7, 7.12, 7.16) and $\mathcal{K}(p)$ collapses to $\mathfrak{q}_p^{\text{univ}}$.

5 The pathion Clifford triad

The structural heart of the universal-kill-set theorem is the following result, which we prove algebraically.

Theorem 5.1 (Pathion Clifford Triad)**.** *Let $p \in \mathcal{P}_{\mathbb{P}}^{\text{aph}}$ with $\dim K_p = 4$. The three operators $L_p, L_{\ell_{16}}, L_{\ell_{16}\cdot p}$ acting on $\mathbb{P}$ satisfy:*

1. *$L_a^\top = -L_a$ for each $a \in \{p, \ell_{16}, \ell_{16}\cdot p\}$ (skew-symmetry).*
2. *$\{L_a, L_b\} = 0$ for distinct a, b in the triad (pairwise anti-commutation).*
3. *$\langle a, b\rangle = 0$ for distinct a, b in the triad (mutual orthogonality).*

Properties (1) and (3) are formal facts at any Cayley–Dickson level; property (2) is the substantive content. Note that the three operators do *not* satisfy the alternative-algebra normalization $L_a^2 = -\|a\|^2 I$, since pathion is non-alternative. They form a Clifford-like triad without scalar squares: generators of an anti-commuting set without a Clifford module structure on $\mathbb{P}$.

The proof of property (2) decomposes into three pairs, of which the first two follow from a single lemma and the third reduces stepwise through sedenion and octonion identities.

5.1 The first two pairs: Lemma A

Lemma 5.2 (Lemma A)**.** *For any $\widetilde{p} = (a, b) \in \mathrm{Im}(\mathbb{P})$ with $a, b \in \mathrm{Im}(\mathbb{S})$, and any $v \in \mathbb{P}$:*

$$[\widetilde{p}, \ell_{16}, v] \;+\; [\ell_{16}, \widetilde{p}, v] \;=\; 0.$$

Proof. Direct expansion using the Cayley–Dickson product (1). Compute $\widetilde{p}\cdot\ell_{16} = (-b, a)$ and $\ell_{16}\cdot\widetilde{p} = (b, -a)$. For $v = (v_1, v_2)$, expand the four products $(\widetilde{p}\cdot\ell_{16})\cdot v$, $\widetilde{p}\cdot(\ell_{16}\cdot v)$, $(\ell_{16}\cdot\widetilde{p})\cdot v$, and $\ell_{16}\cdot(\widetilde{p}\cdot v)$, using $\bar{a} = -a$, $\bar{b} = -b$, and $(xy)^* = \bar{y}\bar{x}$. Both halves of $[\widetilde{p}, \ell_{16}, v] + [\ell_{16}, \widetilde{p}, v]$ reduce to sums of sedenion commutators that cancel termwise:

$$\big([\widetilde{p}, \ell_{16}, v] + [\ell_{16}, \widetilde{p}, v]\big)_1 = \big(-[b, v_1] + [a, \bar{v}_2]\big) + \big([b, v_1] - [a, \bar{v}_2]\big) = 0,$$
$$\big([\widetilde{p}, \ell_{16}, v] + [\ell_{16}, \widetilde{p}, v]\big)_2 = \big([b, v_2] + [a, \bar{v}_1]\big) + \big(-[b, v_2] - [a, \bar{v}_1]\big) = 0. \qquad \square$$

The standard non-associative-algebra identity $\{L_a, L_b\}v = (ab + ba)v - ([a, b, v] + [b, a, v])$ combined with $\widetilde{p}\ell_{16} + \ell_{16}\widetilde{p} = 0$ (direct CD computation) and Lemma 5.2 immediately gives $\{L_{\widetilde{p}}, L_{\ell_{16}}\} = 0$ as operators on $\mathbb{P}$.

Setting $\widetilde{p} = p$ closes the pair (p, ℓ_{16}). Setting $\widetilde{p} = \ell_{16} \cdot p = (s_2, -s_1)$ closes the pair $(\ell_{16} \cdot p, \ell_{16})$.

5.2 The third pair: reduction to a sedenion identity

The remaining pair is $(p, \ell_{16} \cdot p)$. Set $q = \ell_{16} \cdot p$. Algebra-level: $pq + qp = \|p\|^2\ell_{16} - \|p\|^2\ell_{16} = 0$ by direct CD. The associator content is the substantive part.

Proposition 5.3 (Δ-Reduction)**.** *Let $p = (s_1, s_2) \in \mathrm{Im}(\mathbb{P})$ with $s_1, s_2 \in \mathrm{Im}(\mathbb{S})$, and let $q = \ell_{16} \cdot p = (s_2, -s_1)$. Then for every $v = (v_1, v_2) \in \mathbb{P}$,*

$$[p, q, v] + [q, p, v] \;=\; \big(-\Delta(s_1, s_2)\, v_1,\; +\Delta(s_1, s_2)\, v_2\big),$$

where $\Delta(a, b) := \{L_a^{(\mathbb{S})}, L_b^{(\mathbb{S})}\} - \{R_a^{(\mathbb{S})}, R_b^{(\mathbb{S})}\}$ is a sedenion-level operator on $\mathbb{S}$.

Proof. Direct CD expansion of $[p, q, v] + [q, p, v]$ produces four groups of terms. The mixed cross-half terms involving products of the form $s_i(\bar{v}_2 s_i)$ and $(s_i \bar{v}_2)s_i$ cancel in pairs by *sedenion flexibility* $[s, w, s]^{(\mathbb{S})} = 0$, since $\mathbb{S}$ is flexible. The same-half terms reduce, after using $\bar{s}_i = -s_i$, to $L^{(\mathbb{S})}$ and $R^{(\mathbb{S})}$ anti-commutators acting on v_1 and v_2 respectively, with the structure asserted in the proposition. The detailed computation, while tedious, uses no identity beyond flexibility, the imaginarity of s_1, s_2, and CD recursion. □

Theorem 5.4 (Sedenion Δ Identity)**.** *For all $a, b \in \mathrm{Im}(\mathbb{S})$:*

$$\Delta(a, b) := \{L_a^{(\mathbb{S})}, L_b^{(\mathbb{S})}\} - \{R_a^{(\mathbb{S})}, R_b^{(\mathbb{S})}\} = 0 \quad \textit{as operators on } \mathbb{S}.$$

Remark 5.5 (Orthogonality not required)**.** The orthogonality assumption $\langle a, b\rangle = 0$ is unnecessary. The inner-product term $(ab + ba) = -2\langle a, b\rangle$ contributes identically to $\{L_a, L_b\}$ and $\{R_a, R_b\}$, so it cancels in Δ regardless of $\langle a, b\rangle$. Numerical certification (Appendix A) confirms $\Delta = 0$ across both orthogonal and non-orthogonal random imaginary pairs at machine precision.

The proof of Theorem 5.4 uses a four-block CD decomposition that reduces the sedenion identity to a pure octonion identity.

5.3 Four-block decomposition and the Ω identity

For $a = (\alpha, A)$, $b = (\beta, B)$ with $\alpha, \beta \in \mathrm{Im}(\mathbb{O})$ and $A, B \in \mathbb{O}$ (real parts allowed in A, B, since the imaginarity of a in $\mathbb{S}$ requires only $\alpha \in \mathrm{Im}(\mathbb{O})$; the second half A may have a real part as long as the full $a = (\alpha, A) \in \mathrm{Im}(\mathbb{S})$, which here means α imaginary and A unrestricted), bilinearity in (a, b) gives

$$\Delta(a, b) = \Delta_{(\mathrm{i})} + \Delta_{(\mathrm{ii})} + \Delta_{(\mathrm{iii})} + \Delta_{(\mathrm{iv})}$$

where the four blocks correspond to the four CD-decomposition pairs $\big((\alpha, 0), (\beta, 0)\big)$, $\big((\alpha, 0), (0, B)\big)$, $\big((0, A), (\beta, 0)\big)$, and $\big((0, A), (0, B)\big)$ respectively. By the $\Delta(x, y) = \Delta(y, x)$ symmetry, $\Delta_{(\mathrm{iii})} = \Delta_{(\mathrm{ii})}$.

Lemma 5.6 (Four-block analysis)**.**

(*i*) $\Delta_{(i)} = 0$ *by octonion alternativity:* $\{L_\alpha, L_\beta\}_{\mathbb{O}} = \{R_\alpha, R_\beta\}_{\mathbb{O}} = -2\langle\alpha, \beta\rangle I$.

(*ii*) $\Delta_{(ii)} = 0$ *by octonion associator alternation* (*consequence of alternativity*) *combined with the real-vanishing property of* $[\alpha, B, c]_{\mathbb{O}}$ *for* $\alpha \in \mathrm{Im}(\mathbb{O})$.

(*iii*) $\Delta_{(iv)}$ *has trivially vanishing second half; its first half reduces to the octonion identity* Ω *below.*

Theorem 5.7 (Octonion Ω Identity)**.** *For all $A, B, c \in \mathbb{O}$:*

$$\Omega(A, B; c) := \bar{A}(Bc) + \bar{B}(Ac) - (c\bar{B})A - (c\bar{A})B = 0.$$

Proof. Ω is bilinear in (A, B) and symmetric under $A \leftrightarrow B$. By polarization, it suffices to prove the diagonal $\Omega(A, A; c) = 0$.

The diagonal is

$$\Omega(A, A; c) = 2\big[\bar{A}(Ac) - (c\bar{A})A\big].$$

Since $\mathbb{O}$ is alternative, by Artin's theorem the subalgebra $\langle A, c\rangle$ generated by A and c is associative. In particular $\bar{A} = 2\,\mathrm{Re}(A) - A \in \langle A\rangle \subset \langle A, c\rangle$, so:

- $\bar{A}(Ac) = (\bar{A}A)c = \|A\|^2 c$ by associativity in $\langle A, c\rangle$ and the composition norm form $\bar{A}A = \|A\|^2 \in \mathbb{R}$.
- $(c\bar{A})A = c(\bar{A}A) = \|A\|^2 c$ by the same.

Hence $\Omega(A, A; c) = 2\|A\|^2 c - 2\|A\|^2 c = 0$. By polarization $2\Omega(A, B; c) = \Omega(A + B, A + B; c) - \Omega(A, A; c) - \Omega(B, B; c) = 0$, and over $\mathbb{R}$ this gives $\Omega(A, B; c) = 0$. □ □

Proof of Theorem 5.4. By the four-block decomposition and Lemma 5.6, $\Delta(a, b) = 0$ for all imaginary $a, b \in \mathrm{Im}(\mathbb{S})$ if and only if the Ω identity holds in $\mathbb{O}$. Theorem 5.7 provides this. □ □

Proof of Theorem 5.1, completion. Property (2) for the pair $(p, \ell_{16} \cdot p)$ follows by combining $pq + qp = 0$ (direct CD) with Proposition 5.3 and Theorem 5.4. The other two pairs are closed by Lemma 5.2. Properties (1) and (3) are formal. □ □

5.4 The full symbolic chain

The proof chain is summarized as a sequence of implications, each line implying the one above it:

(1) R137 step 6: $[L_a, \pi_p] = 0$ for $a \in \mathfrak{q}_p^{\mathrm{univ}}$.

(2) Centralization with Gram: $[L_a, G_p] = 0$, since $G_p = -L_p^2$ (formal: $L_p^\top = -L_p$) and $\pi_p \in \mathbb{R}[G_p]$.

(3) Anti-commutation: $\{L_a, L_p\} = 0$ (Pathion Clifford Triad, Theorem 5.1).

(4) Two routes to anti-commutation:

- Lemma A closes the pairs (p, ℓ_{16}) and $(\ell_{16} \cdot p, \ell_{16})$ by direct CD computation.
- The pair $(p, \ell_{16} \cdot p)$ reduces to the sedenion identity $\Delta(s_1, s_2) = 0$ for imaginary $s_1, s_2 \in \mathrm{Im}(\mathbb{S})$.

(5) Four-block CD decomposition: $\Delta = 0$ holds iff each of the four blocks vanishes:

- Case (i): octonion alternativity.
- Case (ii): associator alternation in $\mathbb{O}$.
- Case (iv): the octonion identity $\Omega(A, B; c) = 0$.

(6) Octonion Ω theorem: $\Omega = 0$ follows from polarization, Artin's theorem on two-generator subalgebras of $\mathbb{O}$, and the composition norm form $\bar{A}A = \|A\|^2$.

Every node is either a formal Cayley–Dickson identity, a classical alternative-algebra fact (Artin's theorem, octonion alternativity, the composition norm form), or a generic linear-algebra fact (polarization, spectral theorem). No empirical content remains.

6 The $\mathbb{Z}_2$-graded Clifford sublattice

The dim-5 kill set $\mathfrak{q}_p^{(32)}$ at dyadic basepoints decomposes naturally relative to the two-half structure of $p = (s_1, s_2)$.

Definition 6.1. The *symmetric* sublattice is $\mathfrak{q}_p^{\text{univ}} = \text{span}\{p, \ell_{16}, \ell_{16} \cdot p\}$, dimension 3. The *antisymmetric* sublattice is $\mathfrak{q}_p^{\text{anti}} = \text{span}\{(s_1, -s_2), (s_2, s_1)\}$, dimension 2. Together $\mathfrak{q}_p^{(32)} = \mathfrak{q}_p^{\text{univ}} \oplus \mathfrak{q}_p^{\text{anti}}$, dimension 5.

Define the involution $\sigma : \mathbb{P} \to \mathbb{P}$ by

$$\sigma(x, y) = (x, -y) \qquad \text{for } (x, y) \in \mathbb{P} = \mathbb{S} \oplus \mathbb{S} \cdot \ell_{16}.$$

This is the second-half-negation map; it satisfies $\sigma^2 = \text{id}$. The involution acts on the kill set as follows: $\sigma(p) = (s_1, -s_2)$, $\sigma(\ell_{16} \cdot p) = (s_2, s_1)$, $\sigma(\ell_{16}) = -\ell_{16}$. Thus σ exchanges the universal triad with the set $\{(s_1, -s_2), -\ell_{16}, (s_2, s_1)\}$, which spans the same five-dimensional space $\mathfrak{q}_p^{(32)}$ as $\mathfrak{q}_p^{\text{univ}} \cup \mathfrak{q}_p^{\text{anti}}$ — the antisymmetric sublattice $\mathfrak{q}_p^{\text{anti}}$ is precisely the σ-image of $\mathfrak{q}_p^{\text{univ}} \setminus \mathbb{R}\ell_{16}$.

Lemma 6.2 (σ is an algebra automorphism)**.** *The map $\sigma : \mathbb{P} \to \mathbb{P}$, $\sigma(x, y) = (x, -y)$, is an algebra automorphism: $\sigma(z \cdot z') = \sigma(z) \cdot \sigma(z')$ for all $z, z' \in \mathbb{P}$.*

Proof. Direct Cayley–Dickson computation. With $z = (x, y)$ and $z' = (c, d)$:

$$\begin{aligned} \sigma(z \cdot z') &= \sigma(xc - \bar{d}\,y,\ dx + y\bar{c}) = (xc - \bar{d}\,y,\ -dx - y\bar{c}), \\ \sigma(z) \cdot \sigma(z') &= (x, -y)(c, -d) = (xc - \overline{(-d)}\,(-y),\ (-d)x + (-y)\bar{c}) \\ &= (xc - \bar{d}\,y,\ -dx - y\bar{c}). \end{aligned}$$

The two are identical. □ □

Theorem 6.3 ($\mathbb{Z}_2$-Graded Clifford Sublattice)**.** *At every pathion-aphanic basepoint p with $\dim K_p = 4$:*

(1) Distinct elements of $\mathfrak{q}_p^{\text{univ}}$ pairwise anti-commute as left-multiplication operators.

(2) Distinct elements of $\mathfrak{q}_p^{\text{anti}}$ pairwise anti-commute as left-multiplication operators.

(3) Mixed pairs (one from each summand) do not in general anti-commute.

Proof. (1) is the Pathion Clifford Triad Theorem 5.1, already proven algebraically.

(2) follows from (1) by σ-transport. Since σ is an algebra automorphism (Lemma 6.2), left-multiplication operators transport by conjugation:

$$L_{\sigma(a)} \;=\; \sigma \circ L_a \circ \sigma^{-1} \qquad \text{for all } a \in \mathbb{P}. \tag{3}$$

Indeed, $L_{\sigma(a)}(v) = \sigma(a)v = \sigma(a)\sigma(\sigma^{-1}(v)) = \sigma(a \cdot \sigma^{-1}(v)) = (\sigma \circ L_a \circ \sigma^{-1})(v)$ by the automorphism property. Anti-commutators therefore transport as well:

$$\{L_{\sigma(a)}, L_{\sigma(b)}\} \;=\; \sigma \circ \{L_a, L_b\} \circ \sigma^{-1}.$$

In particular $\{L_a, L_b\} = 0$ if and only if $\{L_{\sigma(a)}, L_{\sigma(b)}\} = 0$.

Apply this to the three pairs of the universal triad. Setting $u := \sigma(p) = (s_1, -s_2)$ and $w := \sigma(\ell_{16} \cdot p) = (s_2, s_1)$:

- $\{L_p, L_{\ell_{16}}\} = 0 \implies \{L_u, L_{-\ell_{16}}\} = 0 \implies \{L_u, L_{\ell_{16}}\} = 0.$
- $\{L_p, L_{\ell_{16} \cdot p}\} = 0 \implies \{L_u, L_w\} = 0.$

- $\{L_{\ell_{16}}, L_{\ell_{16}\cdot p}\} = 0 \implies \{L_{\ell_{16}}, L_w\} = 0.$

The second of these is exactly the q_antisym pairwise anti-commutation $\{L_u, L_w\} = 0$. Together with the orthogonality and skew-symmetry of u and w (formal at any CD level), this proves (2).

(3) is an empirical observation, certified at machine zero by direct computation across $N = 8$ random aphanic basepoints (Appendix A); it is not a proved claim. Specifically, $\{L_p, L_w\}$ and $\{L_{\ell_{16}\cdot p}, L_u\}$ have Frobenius norm bounded below by $\Theta(1)$ — a 15-order separation from the within-summand zeros. A symbolic explanation of this asymmetry, in terms of the σ-action on operator pairs, is sketched in Section 8: the failed mixed pairs (p, w) and $(\ell_{16} \cdot p, u)$ are precisely the pairs in which σ swaps elements within the pair (rather than mapping one element to the other globally), so the transport argument does not apply. □ □

Remark 6.4 (Two Clifford triads share ℓ_{16})**.** The universal triad $\{p, \ell_{16}, \ell_{16} \cdot p\}$ and its σ-image $\{u, \ell_{16}, w\}$ are both Clifford triads on $\mathbb{P}$, sharing the generator ℓ_{16}. Together they exhaust the dim-5 space $\mathfrak{q}_p^{(32)}$ as $\mathrm{span}\{p, u\} \oplus \mathbb{R}\ell_{16} \oplus \mathrm{span}\{\ell_{16} \cdot p, w\}$. The failed mixed pairs (p, w) and $(\ell_{16} \cdot p, u)$ are precisely the "cross-triad" pairs that involve elements from one Clifford triad and the σ-conjugate non-corresponding element of the other.

Remark 6.5. The $\mathbb{Z}_2$-grading explains why R136's dim-5 extension does not require the Clifford-triad mechanism with the universal triad. The antisymmetric sublattice $\mathfrak{q}_p^{\mathrm{anti}}$ is independently Clifford-internal (this theorem) but does not anti-commute with the universal triad. The centralization of G_p by elements of $\mathfrak{q}_p^{\mathrm{anti}}$ at dyadic basepoints follows instead from the two-regime closure formalized in §7: a half-localization argument (Lemma 7.6) closes the u-side across all of dyadic; at canonical Stratum-II the w-side closes by block-diagonality of L_w in the G_p-eigenspace decomposition (Theorem 7.13); at the CD-aligned subfamily the w-side closes by direct Clifford-anti-commutation $\{L_w, L_p\} = 0$ (Lemma 7.14). The Lagrange-3 polynomial form $G_p^+ = \frac{7}{4\|p\|^4} G_p - \frac{3}{4\|p\|^6} G_p^2$ remains consistent with the closure but is not the load-bearing element. At generic aphanic the dyadic-only mechanisms fail and $\mathcal{K}(p)$ collapses to $\mathfrak{q}_p^{\mathrm{univ}}$.

7 The two-regime dyadic closure

The dyadic extension Theorem 4.3 carries, in its v1 proof-sketch form, a residual reliance on the Lagrange-3 polynomial representation of G_p^+ as the load-bearing element. The $\mathbb{Z}_2$-grading of §6 sharpens the question: distinct elements of $\mathfrak{q}_p^{\mathrm{anti}}$ pairwise anti-commute (Theorem 6.3 (2), proved by σ-transport from the Pathion Clifford Triad Theorem), but the centralization $[L_a, G_p] = 0$ for $a \in \mathfrak{q}_p^{\mathrm{anti}}$ does not follow from the centralization chain (Lemma 4.2) alone: $\{L_a, L_p\}$ is generically nonzero across the aphanic locus. At dyadic basepoints with three-valued $\mathrm{Spec}(G_p)$, however, the centralization holds at machine precision. The question raised in §8 (Q1) was whether the dyadic centralization admits a purely algebraic derivation, and what algebraic mechanism replaces the failing global anti-commutation.

This section answers both questions. We prove that $[L_a, G_p] = 0$ for every $a \in \mathfrak{q}_p^{\mathrm{anti}}$ at every dyadic pathion-aphanic basepoint, by a two-regime mechanism that splits along the dyadic family. The u-side closes universally across the dyadic family by a half-localization argument (§7.3). The w-side closes at canonical Stratum-II by a block-decomposition argument identifying L_w as block-diagonal in the G_p-eigenspace decomposition (§7.5–§7.7), and at the Cayley–Dickson-aligned subfamily by a direct anti-commutation argument (§7.8). The combined result (Theorem 7.17 in §7.9) upgrades Theorem 4.3's sufficiency from "machine-precision sketch" to "algebraic via two-regime closure", and resolves the open question (Q1) of §8: the load-bearing element is not the Lagrange-3 polynomial but the half-localization / block-diagonality / flex collapse structure catalogued below. The Lagrange-3 form remains consistent with the closure but is not used in the derivation.

7.1 Subfamilies and notation

We work throughout with dyadic pathion-aphanic basepoints $p = \iota_1(s_1) + \iota_2(s_2) \in \mathcal{P}_{\mathbb{P}}^{\text{aph}}$ whose Gram spectrum is three-valued. The canonical subfamily is the Stratum II configuration of Proposition 2.2 with $\dim K_p = 4$; the CD-aligned subfamily carries the Class A signature with $\dim K_p = 12$. We refine the dyadic family into the following dyadic subfamilies; the entire closure of this section concerns these subfamilies and *does not* extend to the eight-valued-spectrum random-aphanic locus.

Definition 7.1 (Dyadic basepoint)**.** A pathion-aphanic p is *dyadic* when each sedenion half s_k ($k = 1, 2$) resides entirely in one octonion half of $\mathbb{S}$, i.e., $s_k \in \iota_1^{\mathbb{S}}(\mathrm{Im}(\mathbb{O})) \cup \iota_2^{\mathbb{S}}(\mathrm{Im}(\mathbb{O}))$, where $\iota_1^{\mathbb{S}}, \iota_2^{\mathbb{S}} : \mathbb{O} \to \mathbb{S}$ denote the sedenion embeddings of the two octonion halves at the previous Cayley–Dickson level. Equivalently, in the engine's basis convention, $p = e_i + e_{16+k}$ for some $i \in \{1, \ldots, 15\}$ and $k \in \{2, \ldots, 15\}$; the subfamily $k \neq i+8$ is canonical Stratum II ($\dim K_p = 4$), while the CD-aligned ray $k = i + 8$ has the Class A spectral signature ($\dim K_p = 12$).

Definition 7.2 (Canonical Stratum-II and CD-aligned subfamilies)**.** Within the dyadic family:

- the *canonical Stratum-II* subfamily is the set of dyadic basepoints with $s_1 \in \iota_1^{\mathbb{S}}(\mathrm{Im}(\mathbb{O}))$ and $s_2 \in \iota_2^{\mathbb{S}}(\mathrm{Im}(\mathbb{O}))$, parameterized as $s_1 = (r, 0)$ and $s_2 = (0, x')$ in the sedenion CD pair, with $r, x' \in \mathrm{Im}(\mathbb{O})$ unit and $\langle r, x' \rangle_{\mathbb{O}} = 0$;
- the *CD-aligned* subfamily is the set of dyadic basepoints satisfying $s_2 = s_1 \cdot \ell_8$, equivalently $s_1 = (r, 0)$ and $s_2 = (0, r)$ for the same $r \in \mathrm{Im}(\mathbb{O})$ unit.

The two subfamilies are disjoint by the orthogonality / equality split: canonical Stratum-II requires $r \neq x'$ (and orthogonal), CD-aligned requires $r = x'$ (i.e., $s_2 = s_1 \cdot \ell_8$).

The canonical Stratum-II subfamily is the generic one in the $e_i + e_{16+k}$ parameterization with $k \neq i + 8$; the CD-aligned subfamily is the single ray of values $k = i + 8$. The Gram spectrum multiplicities are $(\dim E_0, \dim E_{\|p\|^2}, \dim E_{2\|p\|^2}) = (4, 24, 4)$ at canonical Stratum-II and $(12, 8, 12)$ at CD-aligned, both three-valued in $\|p\|^2 \cdot \{0, 1, 2\}$. We write E_λ for the λ-eigenspace of G_p and $P_\lambda : \mathbb{R}^{32} \to E_\lambda$ for the orthogonal projector, so $G_p = \sum_\lambda \lambda P_\lambda$ on the full pathion module.

Definition 7.3 (Antisymmetric kill-set basis)**.** Following §6 we write

$$u \;:=\; \sigma(p) \;=\; (s_1, -s_2), \qquad w \;:=\; \sigma(\ell_{16} \cdot p) \;=\; (s_2, s_1),$$

so $\mathfrak{q}_p^{\text{anti}} = \mathrm{span}\{u, w\}$ and $u, w \in \mathrm{Im}(\mathbb{P})$ as expected. The verification $\ell_{16} \cdot p = (s_2, -s_1)$ (one-line CD-1 calculation) gives $\sigma(\ell_{16} \cdot p) = (s_2, s_1)$ at once.

Definition 7.4 (Half-embedded operators)**.** For $k \in \{1, 2\}$ set

$$A_k \;:=\; L_{\iota_1(s_k)}, \qquad B_k \;:=\; L_{\iota_2(s_k)}.$$

Linearity of $L_{(\cdot)}$ in its argument gives, for any dyadic basepoint:

$$L_p \;=\; A_1 + B_2, \qquad L_u \;=\; A_1 - B_2, \qquad L_w \;=\; A_2 + B_1, \qquad L_{\ell_{16} \cdot p} \;=\; B_1 - A_2.$$

The first and third decompositions are the load-bearing tools for §7.3 and §7.8, respectively; the second pair are useful in cross-checks.

7.2 Primitive identities

The closure of the section uses only the following primitive identities, each formal at a fixed Cayley–Dickson level. We label them for back-reference. All three top-level identities (CD-1, CD-2, N-1) hold at every CD level; the next four (O-LA, O-RA, O-AC, oct-flex) live in the octonions; and the last (sed-flex) lives at sedenion. The four sedenion lemmas L1–L4 are derived from these primitives by a one-line direct sed-CD walk and are used as compact substitutions in the worked sub-cases below.

(CD-1) Cayley–Dickson product. At any level,

$$(a,b)(c,d) \;=\; (ac - \bar{d}\,b,\; da + b\bar{c}). \tag{4}$$

(CD-2) Cayley–Dickson conjugation. At any level,

$$\overline{(a,b)} \;=\; (\bar{a}, -b), \tag{5}$$

hence $\overline{y} = -y$ for any imaginary y at any CD level.

(N-1) Composition norm form on imaginaries. At any CD level, $y\bar{y} = \bar{y}y = \|y\|^2 \in \mathbb{R}_{\geq 0}$; for imaginary y, $y^2 = -\|y\|^2$.

(O-LA) Octonion left-alternativity. For $y, z \in \mathbb{O}$, $y\,(yz) = y^2\,z$.

(O-RA) Octonion right-alternativity. For $y, z \in \mathbb{O}$, $(zy)\,y = z\,y^2$.

(O-AC) Octonion polarization (anti-commutator on imaginaries). For $y, z \in \mathrm{Im}(\mathbb{O})$,

$$yz + zy \;=\; -2\langle y, z\rangle_{\mathbb{O}} \cdot 1. \tag{6}$$

In particular $yz + zy = 0$ when $\langle y, z\rangle = 0$.

(oct-flex) Octonion flexibility. For $y, z \in \mathbb{O}$, $y\,(zy) = (yz)\,y$. Equivalently, by Artin's theorem on two-generator subalgebras of $\mathbb{O}$, the subalgebra $\langle y, z\rangle$ is associative, so any associator with two equal arguments vanishes. This subsumes O-LA and O-RA but we cite the latter where the proof needs only the alternative form.

(sed-flex) Sedenion flexibility. For $a, b \in \mathbb{S}$, $a\,(ba) = (ab)\,a$. Sedenions are flexible but not alternative; two-generator subalgebras are not in general associative, but the flex identity (which is the symmetrized weakening of associativity) survives the Cayley–Dickson doubling from $\mathbb{O}$.

The next four sedenion lemmas package the half-step CD contractions used repeatedly in B1 and B2. We work at canonical Stratum-II with $s_1 = (r, 0)$, $x = (x', 0)$, $s_2 = x \cdot \ell_8 = (0, x')$, $r, x' \in \mathrm{Im}(\mathbb{O})$ unit, $\langle r, x'\rangle_{\mathbb{O}} = 0$. (At CD-aligned, the same lemmas hold with $x' = r$, but the orthogonality hypothesis is replaced by the equality.)

Lemma 7.5 (Sedenion lemmas L1–L4)**.** *At canonical Stratum-II,*

$$\begin{aligned}
&\textit{(L1)} && s_2 \cdot x \;=\; +\ell_8, \\
&\textit{(L2)} && s_2 \cdot \ell_8 \;=\; -x, \\
&\textit{(L3)} && \ell_8 \cdot s_1 \;=\; -\,s_1 \cdot \ell_8, \\
&\textit{(L4)} && (s_1 \cdot \ell_8) \cdot s_1 \;=\; +\ell_8, \qquad s_1 \cdot (s_1 \cdot \ell_8) \;=\; -\ell_8.
\end{aligned}$$

Proof. Each follows from a one-line direct sed-CD walk via (4), (5), and N-1 on the unit imaginary $r, x' \in \mathrm{Im}(\mathbb{O})$. For (L1), $s_2 \cdot x = (0, x')(x', 0) = (0 \cdot x' - \bar{0} \cdot x',\; 0 \cdot 0 + x'\bar{x}') = (0, \|x'\|^2) = (0, 1) = \ell_8$. For (L2), $s_2 \cdot \ell_8 = (0, x')(0, 1) = (0 - \bar{1} \cdot x',\; 1 \cdot 0 + x'\bar{0}) = (-x', 0) = -x$. For (L3), $\ell_8 \cdot s_1 = (0, 1)(r, 0) = (-\bar{0} \cdot 1, 0 + 1 \cdot \bar{r}) = (0, -r)$ and $s_1 \cdot \ell_8 = (r, 0)(0, 1) = (-\bar{1} \cdot 0, 1 \cdot r + 0) = (0, r)$, so $\ell_8 s_1 = -(s_1 \ell_8)$. For (L4), $(s_1\ell_8)s_1 = (0, r)(r, 0) = (0 - \bar{0} \cdot r, 0 \cdot 0 + r\bar{r}) = (0, 1) = \ell_8$, and $s_1(s_1\ell_8) = (r, 0)(0, r) = (-\bar{r} \cdot 0, r \cdot r + 0) = (0, -\|r\|^2) = -\ell_8$. □ □

The pair (L4) is the explicit witness that s_1 does not commute with $s_1\ell_8$ in $\mathbb{S}$ even though both lie in commuting subalgebras $(s_1) \subset \iota_1^{\mathbb{S}}(\mathbb{O})$ and $(s_1\ell_8) \subset \iota_2^{\mathbb{S}}(\mathbb{O})$; this asymmetry is exactly what powers the L_w-action signs ± 2 in B1 below.

7.3 The u-Closure Lemma

Lemma 7.6 (u-Closure Lemma)**.** *At every dyadic pathion-aphanic basepoint $p = \iota_1(s_1) + \iota_2(s_2)$ (in the sense of Definition 7.1), the operators L_u and L_p on $\mathbb{P}$ satisfy $\{L_u, L_p\} = 0$. Consequently, by the centralization chain (Lemma 4.2),*

$$[L_u, G_p] \;=\; 0$$

at every dyadic basepoint, and (the same chain) L_u centralizes the spectral projector π_p as well.

The proof has three steps. Step 1 is the linearity reduction $\{L_u, L_p\} = 2(A_1^2 - B_2^2)$, formal at any CD level. Step 2 is the *half-localization lemma* computing $L^2_{\iota_k(s)} = -\|s\|^2 I$ when s resides in one octonion half of $\mathbb{S}$; we work all four sub-cases (oct-half $\times$ embedding) by direct CD walk on a generic $v = (a, b) \in \mathbb{P}$. Step 3 substitutes the half-localization into the linearity reduction and uses the aphanic equal-norm condition $\|s_1\| = \|s_2\|$ to collapse the residual to zero.

Proof of Lemma 7.6. **Step 1: linearity reduction.** By Definition 7.4, $L_p = A_1 + B_2$ and $L_u = A_1 - B_2$. The anti-commutator-of-sum-and- difference identity (formal at any CD level, no associativity hypothesis on the operators) gives

$$\begin{aligned}\{L_u, L_p\} &= (A_1 - B_2)(A_1 + B_2) + (A_1 + B_2)(A_1 - B_2)\\ &= (A_1^2 + A_1B_2 - B_2A_1 - B_2^2) + (A_1^2 - A_1B_2 + B_2A_1 - B_2^2)\\ &= 2A_1^2 - 2B_2^2.\end{aligned}$$

The cross terms A_1B_2 and B_2A_1 cancel formally; the surviving two terms are scalars (in the sense established below) by the half-localization lemma in Step 2.

Step 2: half-localization lemma. We claim that for any $s \in \mathrm{Im}(\mathbb{S})$ residing entirely in one octonion half of $\mathbb{S}$ (i.e., $s = (s', 0)$ or $s = (0, s')$ at sed-CD pair, with $s' \in \mathrm{Im}(\mathbb{O})$):

$$L^2_{\iota_1(s)} \;=\; -\|s\|^2 I_{\mathbb{P}}, \qquad L^2_{\iota_2(s)} \;=\; -\|s\|^2 I_{\mathbb{P}}, \tag{7}$$

as endomorphisms of $\mathbb{P}$. The proof is a direct two-step pathion-CD walk on a generic $v = (a, b) \in \mathbb{P}$, with $a, b \in \mathbb{S}$ split at sed-CD pair as $a = (a_1, a_2)$, $b = (b_1, b_2)$, $a_k, b_k \in \mathbb{O}$. Note a_k, b_k are *not* required to be imaginary; the CD walk handles general octonion components. We work all four sub-cases.

Sub-case (i): ι_1-embedding, $s = (s', 0)$. By (4) at pathion CD:

$$\iota_1(s) \cdot v \;=\; (s, 0)(a, b) \;=\; (sa - \bar{b} \cdot 0,\; bs + 0 \cdot \bar{a}) \;=\; (sa,\; bs).$$

Iterating once more,

$$\iota_1(s)^2 \cdot v \;=\; (s, 0)(sa,\, bs) \;=\; (s(sa),\; (bs)s).$$

We expand each sed-component via (4) at sed-CD on $s = (s', 0)$ and $a = (a_1, a_2)$:

$$sa \;=\; (s', 0)(a_1, a_2) \;=\; (s'a_1,\; a_2 s'),$$

and on $b = (b_1, b_2)$:

$$bs \;=\; (b_1, b_2)(s', 0) \;=\; (b_1 s',\; -b_2 s'),$$

where the second-half sign uses (5): $\bar{0} = 0$, but the sed-CD form has the $\bar{s'}$ contribution to the second slot, and $\bar{s'} = -s'$ (s' imaginary in $\mathbb{O}$, by N-1). Iterating:

$$\begin{aligned}s(sa) &= (s', 0)(s'a_1,\, a_2 s')\\ &= \big(s'(s'a_1) - \overline{a_2 s'} \cdot 0,\; (a_2 s')s' + 0 \cdot \overline{s'a_1}\big)\\ &= \big(s'(s'a_1),\; (a_2 s')s'\big),\end{aligned}$$

$$\begin{aligned}(bs)s &= (b_1 s',\, -b_2 s')(s', 0)\\ &= \big((b_1 s')s' - \bar{0}\cdot(-b_2 s'),\; 0\cdot(b_1 s') + (-b_2 s')\cdot\overline{s'}\big)\\ &= \big((b_1 s')s',\; (-b_2 s')(-s')\big) \;=\; \big((b_1 s')s',\; (b_2 s')s'\big).\end{aligned}$$

The four octonion-slot products collapse by O-LA / O-RA / N-1:

$$\begin{aligned} s'(s'a_1) &\overset{\text{O-LA}}{=} (s')^2 a_1 \overset{\text{N-1}}{=} -\|s'\|^2 a_1, & (a_2 s')s' &\overset{\text{O-RA}}{=} a_2 (s')^2 \overset{\text{N-1}}{=} -\|s'\|^2 a_2,\\ (b_1 s')s' &\overset{\text{O-RA}}{=} b_1 (s')^2 \overset{\text{N-1}}{=} -\|s'\|^2 b_1, & (b_2 s')s' &\overset{\text{O-RA}}{=} b_2 (s')^2 \overset{\text{N-1}}{=} -\|s'\|^2 b_2.\end{aligned}$$

Reassembling:

$$\iota_1(s)^2 \cdot v \;=\; \big((-\|s'\|^2 a_1, -\|s'\|^2 a_2),\; (-\|s'\|^2 b_1, -\|s'\|^2 b_2)\big) \;=\; -\|s'\|^2 v \;=\; -\|s\|^2 v,$$

using $\|s\| = \|s'\|$ when $s = (s', 0) \in \mathbb{S}$. ✓

Sub-case (ii): ι_1-embedding, $s = (0, s')$. The pathion expansion is the same, $\iota_1(s)^2 v = (s(sa), (bs)s)$, but the sed-CD products differ. With $s = (0, s')$, $a = (a_1, a_2)$:

$$sa \;=\; (0, s')(a_1, a_2) \;=\; (-\bar{a_2} s',\; s'\bar{a_1}), \qquad bs \;=\; (b_1, b_2)(0, s') \;=\; (-\bar{s'} b_2,\; s' b_1) \;=\; (s' b_2,\; s' b_1),$$

where the last equality uses $\bar{s'} = -s'$. Iterating:

$$\begin{aligned} s(sa) &= (0, s')(-\bar{a_2} s',\; s'\bar{a_1})\\ &= \big(-\overline{s'\bar{a_1}}\cdot s',\; s'\cdot\overline{-\bar{a_2} s'}\big),\end{aligned}$$

where the first slot uses $0\cdot(-\bar{a_2}s') = 0$ and the $-\bar{d}\cdot b = -\overline{s'\bar{a_1}}\cdot s'$ term, and the second slot uses $(s'\bar{a_1})\cdot 0 = 0$ and the $b\bar{c} = s'\cdot\overline{-\bar{a_2}s'}$ term. The conjugates simplify:

$$\overline{s'\bar{a_1}} \;=\; (\bar{a_1})^*(s')^* \;=\; a_1\cdot(-s') \;=\; -a_1 s', \qquad \overline{-\bar{a_2}s'} \;=\; -((s')^*(\bar{a_2})^*) \;=\; -(\bar{s'}a_2) \;=\; s'a_2.$$

Substituting:

$$s(sa) \;=\; \big(-(-a_1 s')s',\; s'\cdot s'a_2\big) \;=\; \big((a_1 s')s',\; (s')^2 a_2\big) \;=\; (-\|s'\|^2 a_1,\; -\|s'\|^2 a_2),$$

the last equality by O-RA on $(a_1 s')s' = a_1 (s')^2$ and the scalar $(s')^2 a_2$ via N-1 (no O-LA needed since the scalar $(s')^2 \in \mathbb{R} \subset \mathbb{O}$ centralizes a_2). Similarly,

$$\begin{aligned}(bs)s &= (s' b_2,\; s' b_1)(0, s')\\ &= \big(-\bar{s'}\cdot s' b_1,\; (s')^2 b_2\big)\\ &= \big(s'(s' b_1),\; (s')^2 b_2\big) \;=\; (-\|s'\|^2 b_1, -\|s'\|^2 b_2),\end{aligned}$$

by O-LA on the first slot ($s'(s'b_1) = (s')^2 b_1$) and N-1 on the second. Reassembling: $\iota_1(s)^2 v = (-\|s'\|^2 a, -\|s'\|^2 b) = -\|s\|^2 v$. ✓

Sub-case (iii): ι_2-embedding, $s = (s', 0)$. By (4):

$$\iota_2(s)\cdot v \;=\; (0, s)(a, b) \;=\; (-\bar{b}s,\; s\bar{a}).$$

Iterating once,

$$\begin{aligned}\iota_2(s)^2 \cdot v &= (0, s)(-\bar{b}s,\; s\bar{a})\\ &= \big(-\overline{s\bar{a}}\cdot s,\; s\cdot\overline{-\bar{b}s}\big)\\ &= \big(-(a\bar{s})s,\; s(\bar{s}b)\big) \;=\; \big((as)s,\; s(sb)\big),\end{aligned}$$

the last equality using $\bar{s} = -s$ for imaginary $s \in \mathrm{Im}(\mathbb{S})$, hence $-(a\bar{s})s = -(a(-s))s = (as)s$ and $s(\bar{s}b) = s(-sb) = -s(sb)$, which corrects to $+s(sb)$ after matching against the $-\bar{d}b$ sign in (4)

(both signs land on the same final form when carefully tracked through the conjugations). A direct re-derivation: at sed level with $s = (s', 0)$, $a = (a_1, a_2)$,

$$as \;=\; (a_1, a_2)(s', 0) \;=\; (a_1 s',\; -a_2 s'),$$

$$(as)s \;=\; (a_1 s', -a_2 s')(s', 0) \;=\; \big((a_1 s')s',\; (-a_2 s')(-s')\big) \;=\; \big((a_1 s')s',\; (a_2 s')s'\big),$$

and by O-RA on each slot, $(as)s = (-\|s'\|^2 a_1, -\|s'\|^2 a_2) = -\|s'\|^2 a$. Similarly $sb = (s', 0)(b_1, b_2) = (s'b_1, b_2 s')$ at sed-CD, and

$$s(sb) \;=\; (s', 0)(s'b_1,\; b_2 s') \;=\; \big(s'(s'b_1),\; (b_2 s')s'\big) \;=\; (-\|s'\|^2 b_1, -\|s'\|^2 b_2)$$

by O-LA on the first slot and O-RA on the second. Reassembling: $\iota_2(s)^2 v = ((as)s, s(sb)) = (-\|s\|^2 a, -\|s\|^2 b) = -\|s\|^2 v$. ✓

Sub-case (iv): ι_2-embedding, $s = (0, s')$. The same template with $s = (0, s')$. At sed-CD,

$$as \;=\; (a_1, a_2)(0, s') \;=\; (-\bar{s'} a_2,\; s' a_1) \;=\; (s' a_2,\; s' a_1),$$

$$(as)s \;=\; (s' a_2, s' a_1)(0, s') \;=\; \big(-\bar{s'} \cdot s' a_1,\; (s')^2 a_2\big) \;=\; \big(s'(s' a_1),\; (s')^2 a_2\big),$$

using $-\bar{s'} = s'$ and the second-slot $b\bar{c}$ term reducing directly. By O-LA on the first slot and N-1 on the second, $(as)s = (-\|s'\|^2 a_1, -\|s'\|^2 a_2) = -\|s'\|^2 a$. Similarly,

$$sb \;=\; (0, s')(b_1, b_2) \;=\; (-\bar{b_2} s',\; s' \bar{b_1}),$$

$$s(sb) \;=\; (0, s')(-\bar{b_2} s',\; s' \bar{b_1}) \;=\; \big(-\overline{s' \bar{b_1}} \cdot s',\; s' \cdot \overline{-\bar{b_2} s'}\big) \;=\; \big((b_1 s')s',\; (s')^2 b_2\big),$$

using the same conjugation-collapse as in sub-case (ii). By O-RA on the first slot and N-1 on the second, $s(sb) = (-\|s'\|^2 b_1, -\|s'\|^2 b_2) = -\|s'\|^2 b$. Reassembling: $\iota_2(s)^2 v = -\|s\|^2 v$. ✓

In all four sub-cases (7) holds, completing the half-localization lemma.

Step 3: aphanic norm cancellation. Substituting (7) into Step 1's reduction (with $s_1 = (s_1', 0)$ or $(0, s_1')$ in one octonion half by the dyadic hypothesis, and similarly s_2):

$$\{L_u, L_p\} \;=\; 2A_1^2 - 2B_2^2 \;=\; 2\big(-\|s_1\|^2 I_{\mathbb{P}}\big) - 2\big(-\|s_2\|^2 I_{\mathbb{P}}\big) \;=\; 2\big(\|s_2\|^2 - \|s_1\|^2\big) I_{\mathbb{P}}.$$

The aphanic characterization Proposition 2.2 gives $\|s_1\| = \|s_2\|$, hence $\{L_u, L_p\} = 0$.

Centralization chain. By Lemma 4.2, $\{L_u, L_p\} = 0 \Longrightarrow [L_u, L_p^2] = 0 \Longleftrightarrow [L_u, G_p] = 0 \Longrightarrow [L_u, \pi_p] = 0$, which is the second assertion. □ □

Remark 7.7 (Sharpness of the dyadic restriction)**.** The half-localization lemma (7) fails at random aphanic. If $s_1 = (\alpha, \beta)$ has nonzero components in both octonion halves of $\mathbb{S}$ (i.e., $\alpha, \beta \in \mathbb{O}$ with α, β not both zero in $\mathrm{Im}(\mathbb{O})$ part), then the second iteration of $\iota_1(s_1)\cdot$ on a generic $v = (a, b) \in \mathbb{P}$ produces, in the first sed-half slot of the result, a residual three-octonion associator

$$-\big[\bar{\alpha},\; \bar{a_2},\; \beta\big]_{\mathbb{O}},$$

arising from a sed-CD walk in which the cross-product $\bar{\beta} a_2$ appears alongside αa_1. The associator is nonzero for generic $\alpha, \beta, a_2 \in \mathbb{O}$ (octonions are alternative but not associative; three independent imaginary octonions span a non-associative subalgebra). Consequently $L^2_{\iota_1(s_1)} \neq -\|s_1\|^2 I_{\mathbb{P}}$ at random aphanic, the linearity reduction's two squared terms no longer cancel, and $\{L_u, L_p\}$ is generically nonzero, in agreement with diagnostic v2's empirical observation of a Frobenius residual of order $\Theta(1)$ for $\{L_u, L_p\}$ at random aphanic. The dyadic restriction is therefore sharp: each s_k residing in one octonion half is exactly the condition under which the cross-half octonion associator vanishes.

7.4 The recovered kernel basis at canonical Stratum-II

For the w-side closure at canonical Stratum-II we work with an explicit basis of $\ker L_p$. The basis was recovered numerically at canonical Stratum-II by a brute-force enumeration over two-coordinate pathion vectors at $p_0 = e_1 + e_{26}$ and a four-point family check; it then admits a closed-form structural description in terms of the canonical Stratum-II parameters (s_1, x).

Lemma 7.8 (Recovered Kernel Basis)**.** *Let $p = \iota_1(s_1) + \iota_2(s_2)$ be a canonical Stratum-II basepoint with $s_1 = (r, 0)$, $x = (x', 0)$, $s_2 = x \cdot \ell_8 = (0, x')$, $r, x' \in \mathrm{Im}(\mathbb{O})$ unit, and $\langle r, x' \rangle_{\mathbb{O}} = 0$. Define*

$$v_0 = (x,\ s_1 \cdot \ell_8), \qquad v_1 = (s_1 \cdot x,\ \ell_8), \qquad v_2 = \ell_{16} \cdot v_1, \qquad v_3 = \ell_{16} \cdot v_0, \tag{8}$$

all elements of $\mathbb{P}$. Then

(i) $\{v_0, v_1, v_2, v_3\} \subset \ker L_p$, *and*

(ii) $\mathrm{span}_{\mathbb{R}}\{v_0, v_1, v_2, v_3\} = \ker L_p$.

The closed-form expressions for v_2, v_3 unfold via (4). Using $\ell_{16} = (0, 1) \in \mathbb{P}$ and the imaginarity of the second component:

$$v_2 = (0,1)(s_1 x, \ell_8) = (-\bar{\ell_8} \cdot 1,\ \ell_8 \cdot 0 + 1 \cdot \overline{s_1 x}) = (\ell_8,\ -s_1 x),$$

since $\bar{\ell_8} = -\ell_8$ at sed (CD-2, ℓ_8 imaginary in $\mathbb{S}$) and $\overline{s_1 x} = -(s_1 x)$ at sed since $s_1 x = (rx', 0)$ has $rx' \in \mathrm{Im}(\mathbb{O})$ (orthogonal imaginary octonions multiply to an imaginary by O-AC: $rx' + x'r = -2\langle r, x' \rangle = 0$, so $\mathrm{Re}(rx') = \frac{1}{2}(rx' + \overline{rx'}) = \frac{1}{2}(rx' + x'r) = 0$). Likewise

$$v_3 = \ell_{16} \cdot v_0 = (0,1)(x, s_1 \ell_8) = (-\overline{s_1 \ell_8},\ \bar{x}) = (s_1 \ell_8,\ -x),$$

using $\overline{s_1 \ell_8} = -(s_1 \ell_8)$ (sed-imaginary, by direct computation $(s_1 \ell_8)^* = (0, r)^* = (0, -r) = -s_1 \ell_8$ via (5)) and $\bar{x} = -x$ (sed-imaginary).

Proof of (i), partial. For $k = 0, 1$ the kernel inclusions $p \cdot v_k = 0$ are direct CD walks worked in Lemma 7.9 below. For $k = 2, 3$ the kernel inclusions follow from the Pathion Clifford Triad (Theorem 5.1): the anti-commutation $\{L_p, L_{\ell_{16}}\} = 0$ gives $L_p L_{\ell_{16}} = -L_{\ell_{16}} L_p$, hence $p \cdot v_2 = L_p L_{\ell_{16}} v_1 = -L_{\ell_{16}} L_p v_1 = -L_{\ell_{16}} \cdot 0 = 0$, and similarly $p \cdot v_3 = 0$ from $p \cdot v_0 = 0$. □ □

The spanning property (ii) is, in this artifact, taken as the basis definition: numerical certification at the engine level (Section A, post-v8 family check, span residual $\leq 1.57 \times 10^{-16}$) verifies that the four explicit vectors in (8) span the same four-dimensional subspace of $\mathbb{P}$ as the eigenvalue-decomposition basis returned by the engine's $E_0(G_p)$ routine. A purely algebraic proof of (ii) would require showing that the four vectors are linearly independent and exhaust $\ker L_p$; the linear-independence half follows from inspecting the explicit pathion-coordinate forms (no two of v_0, v_1, v_2, v_3 have overlapping support in the engine basis at p_0, hence span dim 4), while the dim-count is supplied by the four-dimensional kernel guaranteed at every Stratum II basepoint (Proposition 2.2).

7.5 B1: Kernel preservation

Lemma 7.9 (B1 – Kernel Preservation)**.** *At every canonical Stratum-II basepoint, L_w preserves $\ker L_p$: $L_w \cdot \ker L_p \subseteq \ker L_p$.*

By Lemma 7.8 (ii) and linearity of L_w, it suffices to verify the four identities $w \cdot v_k \in \ker L_p$ for $k = 0, 1, 2, 3$, or equivalently that each $w \cdot v_k$ is a scalar multiple of one of v_0, v_1, v_2, v_3. We supply explicit L_w-action identities of this form: $w \cdot v_0 = +2v_2$, $w \cdot v_1 = -2v_3$, $w \cdot v_2 = -2v_0$, $w \cdot v_3 = +2v_1$. Together they give L_w as a ± 2 skew signed permutation on the recovered basis with $M_w^2 = -4I$, matching the diagnostic v7/v8 numerical output.

Proof of Lemma 7.9. We work with $p = (s_1, s_2)$, $w = (s_2, s_1)$, $s_1 = (r, 0)$, $x = (x', 0)$, $s_2 = (0, x')$, and the recovered basis from Lemma 7.8. Computations are organized as four kernel checks ($p \cdot v_k = 0$) followed by four L_w-action checks ($w \cdot v_k = c_k v_{\pi(k)}$); the second pair of each four follows from the first by anti-commutation with $L_{\ell_{16}}$.

Kernel check $k = 0$: $p \cdot v_0 = 0$. By (4):

$$p \cdot v_0 \;=\; (s_1, s_2)(x, s_1\ell_8) \;=\; (s_1 x - \overline{s_1\ell_8}\, s_2,\; (s_1\ell_8)s_1 + s_2\bar{x}).$$

First half: $\overline{s_1\ell_8} = -(s_1\ell_8)$ (CD-2 + sed-imag of $s_1\ell_8 = (0, r)$, which has zero first component and imaginary r in the second). So $-\overline{s_1\ell_8}\, s_2 = (s_1\ell_8)s_2$. By direct sed-CD: $(s_1\ell_8)s_2 = (0, r)(0, x') = (-\bar{x}'r, 0) = (x'r, 0) = xs_1$ (at sed-pair, where $xs_1 = (x', 0)(r, 0) = (x'r, 0)$). Combined with $s_1 x = (r, 0)(x', 0) = (rx', 0)$:

$$\text{first half} \;=\; (rx', 0) + (x'r, 0) \;=\; (rx' + x'r,\, 0) \overset{\text{O-AC + Hyp}}{=} (0, 0) \;=\; 0.$$

Second half: $\bar{x} = -x$ (sed-imag), so $s_2\bar{x} = -s_2 x$. By L1, $s_2 x = \ell_8$. By L4, $(s_1\ell_8)s_1 = \ell_8$. So second half $= \ell_8 - \ell_8 = 0$. Combining, $p \cdot v_0 = 0$. ✓

Kernel check $k = 1$: $p \cdot v_1 = 0$. By (4):

$$p \cdot v_1 \;=\; (s_1, s_2)(s_1 x, \ell_8) \;=\; (s_1(s_1 x) - \bar{\ell_8}\, s_2,\; \ell_8 s_1 + s_2\overline{s_1 x}).$$

First half: $\bar{\ell_8} = -\ell_8$, so $-\bar{\ell_8}\, s_2 = \ell_8 s_2$. By direct sed-CD, $\ell_8 s_2 = (0, 1)(0, x') = (-\bar{x}', 0) = (x', 0) = x$. For $s_1(s_1 x)$: at sed, $s_1 = (r, 0)$ and $s_1 x = (rx', 0)$, both in the first oct half. Sed-CD gives $s_1(s_1 x) = (r, 0)(rx', 0) = (r(rx'), 0)$. By O-LA on $\langle r \rangle$ (automatic, single generator): $r(rx') = r^2 x' = -\|r\|^2 x' = -x'$. So $s_1(s_1 x) = (-x', 0) = -x$. Hence first half $= -x + x = 0$. *Second half:* $\overline{s_1 x} = -(s_1 x)$ (sed-imag), so $s_2\overline{s_1 x} = -s_2(s_1 x)$. By L3, $\ell_8 s_1 = -(s_1\ell_8)$, so the second-half sum is $-(s_1\ell_8) - s_2(s_1 x)$. We compute $s_2(s_1 x)$ at sed: $s_2 = (0, x')$, $s_1 x = (rx', 0)$, so

$$s_2(s_1 x) \;=\; (0, x')(rx', 0) \;=\; (-\bar{0} \cdot x',\; 0 \cdot 0 + x' \cdot \overline{rx'}) \;=\; (0,\; x'\overline{rx'}).$$

With rx' imaginary in $\mathbb{O}$, $\overline{rx'} = -(rx')$, so $x'\overline{rx'} = -x'(rx')$. Apply oct-flex on $\langle r, x' \rangle$: $x'(rx') = (x'r)x' \overset{\text{O-AC + Hyp}}{=} (-rx')x' \overset{\text{O-RA}}{=} -r(x')^2 = r$ (using $\langle r, x' \rangle = 0 \Rightarrow x'r = -rx'$, then O-RA on $\langle x', r \rangle$, then $(x')^2 = -1$). Therefore $x'(rx') = r$, and $x'\overline{rx'} = -r$, giving $s_2(s_1 x) = (0, -r) = -(s_1\ell_8)$. So second half $= -(s_1\ell_8) - (-(s_1\ell_8)) = 0$. Combining, $p \cdot v_1 = 0$. ✓

Kernel checks $k = 2, 3$: $p \cdot v_2 = p \cdot v_3 = 0$. By the Pathion Clifford Triad (Theorem 5.1), $\{L_p, L_{\ell_{16}}\} = 0$, hence $L_p L_{\ell_{16}} = -L_{\ell_{16}} L_p$. Applied to v_1 and v_0:

$$p \cdot v_2 \;=\; L_p L_{\ell_{16}} v_1 \;=\; -L_{\ell_{16}} L_p v_1 \;=\; -L_{\ell_{16}} \cdot 0 \;=\; 0,$$

and similarly $p \cdot v_3 = -L_{\ell_{16}} L_p v_0 = 0$. ✓

L_w-action $k = 0$: $w \cdot v_0 = +2v_2$. By (4):

$$w \cdot v_0 \;=\; (s_2, s_1)(x, s_1\ell_8) \;=\; (s_2 x - \overline{s_1\ell_8}\, s_1,\; (s_1\ell_8)s_2 + s_1\bar{x}).$$

First half: $-\overline{s_1\ell_8}\, s_1 = (s_1\ell_8)s_1 = \ell_8$ by L4. By L1, $s_2 x = \ell_8$. So first half $= \ell_8 + \ell_8 = 2\ell_8$. *Second half:* $\bar{x} = -x$, so $s_1\bar{x} = -s_1 x = -(rx', 0)$. By the sed-CD walk above, $(s_1\ell_8)s_2 = (x'r, 0) = xs_1$. So second half $= (x'r, 0) - (rx', 0) = (x'r - rx',\, 0)$. By O-AC + Hyp, $x'r + rx' = 0$, hence $x'r - rx' = -2(rx')$. At the sed-pair level: second half $= -2(rx', 0) = -2(s_1 x)$. Combining, $w \cdot v_0 = (2\ell_8, -2(s_1 x)) = 2(\ell_8, -(s_1 x)) = 2v_2$ (recalling $v_2 = (\ell_8, -s_1 x)$ by the unfold above). ✓

L_w-action $k = 1$: $w \cdot v_1 = -2v_3$. By (4):

$$w \cdot v_1 \;=\; (s_2, s_1)(s_1 x, \ell_8) \;=\; (s_2(s_1 x) - \bar{\ell_8}\, s_1,\; \ell_8 s_2 + s_1\overline{s_1 x}).$$

First half: $-\bar{\ell}_8\, s_1 = \ell_8 s_1 = -(s_1\ell_8)$ by L3. $s_2(s_1 x) = -(s_1\ell_8)$ by the sed-CD walk in the $k=1$ kernel check. So first half $= -(s_1\ell_8) - (s_1\ell_8) = -2(s_1\ell_8)$. *Second half:* $\overline{s_1 x} = -(s_1 x)$, so $s_1\overline{s_1 x} = -s_1(s_1 x) = -(-x) = x$ by the same O-LA derivation as in $p \cdot v_1$. By the $\ell_8 s_2 = x$ identity (sed-CD as in $p \cdot v_1$): second half $= x + x = 2x$. Combining, $w \cdot v_1 = (-2(s_1\ell_8), 2x) = -2(s_1\ell_8, -x) = -2v_3$ (since $v_3 = (s_1\ell_8, -x)$ by the unfold of $\ell_{16} \cdot v_0$ above). ✓

L_w-actions $k = 2, 3$: $w \cdot v_2 = -2v_0$, $w \cdot v_3 = +2v_1$. By Theorem 6.3 (2)'s pair $\{L_{\ell_{16}}, L_w\} = 0$ (which follows from $\{L_{\ell_{16}}, L_{\ell_{16}\cdot p}\} = 0$ by σ-transport in §6), $L_w L_{\ell_{16}} = -L_{\ell_{16}} L_w$. Applied to v_1:

$$w \cdot v_2 \;=\; L_w L_{\ell_{16}} v_1 \;=\; -L_{\ell_{16}} L_w v_1 \;=\; -L_{\ell_{16}}(-2v_3) \;=\; 2\ell_{16} v_3 \;=\; 2\ell_{16}(\ell_{16} v_0) \;=\; -2v_0,$$

using $\ell_{16}^2 = -1$ in $\mathbb{P}$ (CD-2 norm form: $\ell_{16}^2 = -\|\ell_{16}\|^2 = -1$). Similarly, $w \cdot v_3 = L_w L_{\ell_{16}} v_0 = -L_{\ell_{16}} L_w v_0 = -L_{\ell_{16}}(2v_2) = -2\ell_{16} v_2 = -2\ell_{16}(\ell_{16} v_1) = 2v_1$. ✓

In all four cases $w \cdot v_k$ is a scalar multiple of an element of $\{v_0, v_1, v_2, v_3\} \subset \ker L_p$, hence $w \cdot v_k \in \ker L_p$. By Lemma 7.8 (ii) and linearity, $L_w \cdot \ker L_p \subseteq \ker L_p$. □ □

Corollary 7.10 (Kernel associator identity)**.** *For $v \in \ker L_p$,*

$$[w, p, v] \;+\; [p, w, v] \;=\; 0.$$

Proof. By the canonical Stratum-II orthogonality $\langle p, w\rangle_{\mathbb{P}} = 0$ and the polarization identity $ab + ba = -2\langle a, b\rangle \cdot 1$ for $a, b \in \mathrm{Im}(\mathbb{P})$ (which holds at any CD level), we have $pw + wp = 0$ in $\mathbb{P}$. Together with the associator definitions $[a, b, v] = (ab)v - a(bv)$:

$$\begin{aligned}
[w, p, v] \;+\; [p, w, v] &= \big((wp)v - w(pv)\big) + \big((pw)v - p(wv)\big) \\
&= \big((wp) + (pw)\big)v - w(pv) - p(wv) \\
&= 0 - w \cdot 0 - p \cdot (wv) \\
&= -p \cdot (w \cdot v),
\end{aligned}$$

using $pw + wp = 0$ and $pv = 0$. By Lemma 7.9, $w \cdot v \in \ker L_p$, so $p \cdot (w \cdot v) = 0$, giving $[w, p, v] + [p, w, v] = 0$. □ □

The orthogonality $\langle p, w\rangle = 0$ used above is formal at any CD level: $p = (s_1, s_2)$ and $w = (s_2, s_1)$ give $\langle p, w\rangle_{\mathbb{P}} = \langle s_1, s_2\rangle_{\mathbb{S}} + \langle s_2, s_1\rangle_{\mathbb{S}} = 2\langle s_1, s_2\rangle_{\mathbb{S}} = 0$ by the Stratum II hypothesis $\langle s_1, s_2\rangle_{\mathbb{S}} = 0$.

7.6 B2: Top-eigenspace annihilation

Lemma 7.11 (B2 – Top-Eigenspace Annihilation)**.** *At every canonical Stratum-II basepoint, L_w annihilates the top G_p-eigenspace:*

$$L_w \cdot E_{2\|p\|^2}(G_p) \;=\; 0.$$

The proof has three structural steps. Step 1 identifies the top G_p-eigenspace with the σ-image of $\ker L_p$: $E_{2\|p\|^2}(G_p) = \sigma(\ker L_p)$. Step 2 transports the operator L_w across the σ-conjugation to obtain $L_w \circ \sigma = \sigma \circ L_{\ell_{16}\cdot p}$. Step 3 establishes the q_universal annihilation $L_{\ell_{16}\cdot p} \cdot \ker L_p = 0$ on the recovered basis. Combining these gives, for every $v \in \ker L_p$,

$$L_w \cdot \sigma(v) \;=\; \sigma\big(L_{\ell_{16}\cdot p} \cdot v\big) \;=\; \sigma(0) \;=\; 0,$$

which by Step 1 + linearity is exactly the lemma.

Proof of Lemma 7.11. **Step 1: $E_{2\|p\|^2}(G_p) = \sigma(\ker L_p)$.** The pathion automorphism $\sigma(x, y) = (x, -y)$ from Lemma 6.2 satisfies $\sigma^2 = \mathrm{id}$ and is an isometry. Operator transport (3) gives $L_{\sigma(a)} = \sigma \circ L_a \circ \sigma^{-1}$ for every $a \in \mathbb{P}$, hence

$$\sigma \circ L_p \;=\; L_{\sigma(p)} \circ \sigma \;=\; L_u \circ \sigma,$$

which on kernels gives $\sigma(\ker L_p) = \ker L_u$ (a 4-dimensional subspace of $\mathbb{P}$, since u is also a pathion-aphanic basepoint inheriting $\dim\ker L_u = 4$ via σ-symmetry of the aphanic locus).

We now show $\ker L_u \subseteq E_{2\|p\|^2}(G_p)$. From the half-localization lemma (7) applied to both $\iota_1(s_1)$ and $\iota_2(s_2)$ at the dyadic basepoint, $A_1^2 = -\|s_1\|^2 I_{\mathbb{P}}$ and $B_2^2 = -\|s_2\|^2 I_{\mathbb{P}}$. Using $L_p = A_1 + B_2$ and $L_u = A_1 - B_2$ (Definition 7.4):

$$\begin{aligned} L_p^2 + L_u^2 &= (A_1 + B_2)^2 + (A_1 - B_2)^2 \\ &= 2A_1^2 + 2B_2^2 \\ &= -2\|s_1\|^2 I_{\mathbb{P}} - 2\|s_2\|^2 I_{\mathbb{P}} \\ &= -2\|p\|^2 I_{\mathbb{P}}, \end{aligned}$$

where the last equality uses $\|p\|^2 = \|s_1\|^2 + \|s_2\|^2$ (formal at any CD level by the norm form on the pair (s_1, s_2)). For any $v \in \ker L_u$, $L_u^2 v = 0$, hence $L_p^2 v = -2\|p\|^2 v$, which by $G_p = -L_p^2$ on imaginary p (Lemma 4.2) gives $G_p v = 2\|p\|^2 v$. Therefore $v \in E_{2\|p\|^2}(G_p)$, and $\ker L_u \subseteq E_{2\|p\|^2}(G_p)$.

The opposite inclusion follows by dimension count: at canonical Stratum-II, $E_{2\|p\|^2}(G_p)$ has dimension 4 (Definition 7.2), matching $\dim\ker L_u = 4$. Hence

$$\sigma(\ker L_p) \;=\; \ker L_u \;=\; E_{2\|p\|^2}(G_p).$$

Step 2: σ-transport of L_w. By Lemma 6.2 the involution σ is an algebra automorphism, so operator transport (3) gives

$$L_w \circ \sigma \;=\; L_{\sigma(\sigma(w))} \circ \sigma \;=\; \sigma \circ L_{\sigma(w)}.$$

We compute $\sigma(w)$ via (4): $w = (s_2, s_1)$, hence $\sigma(w) = (s_2, -s_1)$. By a one-line CD-1 computation, $\ell_{16} \cdot p = (0, 1)(s_1, s_2) = (-\bar{s_2}, \bar{s_1}) = (s_2, -s_1)$ (using $\bar{s_k} = -s_k$ on imaginaries), so $\sigma(w) = \ell_{16} \cdot p$. Therefore

$$L_w \circ \sigma \;=\; \sigma \circ L_{\ell_{16} \cdot p}. \tag{9}$$

Step 3: q_universal annihilation $L_{\ell_{16}\cdot p} \cdot \ker L_p = 0$. By Lemma 7.8 (ii) and linearity of $L_{\ell_{16}\cdot p}$, it suffices to verify $L_{\ell_{16}\cdot p} \cdot v_k = 0$ for $k = 0, 1, 2, 3$. We work the four cases.

$k = 0$: $L_{\ell_{16}\cdot p} \cdot v_0 = 0$. By (4):

$$L_{\ell_{16}\cdot p} \cdot v_0 \;=\; (s_2, -s_1)(x, s_1\ell_8) \;=\; (s_2 x - \overline{s_1\ell_8}\,(-s_1),\ (s_1\ell_8)s_2 + (-s_1)\bar{x}).$$

First half: $-\overline{s_1\ell_8}\,(-s_1) = \overline{s_1\ell_8}\,s_1 = -(s_1\ell_8)s_1 = -\ell_8$ by L4. By L1, $s_2 x = \ell_8$. Hence first half $= \ell_8 - \ell_8 = 0$. *Second half:* $\bar{x} = -x$, so $(-s_1)\bar{x} = s_1 x = (rx', 0)$. By the sed-CD walk in $p \cdot v_0$, $(s_1\ell_8)s_2 = (x'r, 0)$. Hence second half $= (x'r, 0) + (rx', 0) = (x'r + rx',\ 0) \overset{\text{O-AC + Hyp}}{=} 0$. Combining, $L_{\ell_{16}\cdot p} \cdot v_0 = 0$. ✓

$k = 1$: $L_{\ell_{16}\cdot p} \cdot v_1 = 0$. By (4):

$$L_{\ell_{16}\cdot p} \cdot v_1 \;=\; (s_2, -s_1)(s_1 x, \ell_8) \;=\; (s_2(s_1 x) - \bar{\ell_8}\,(-s_1),\ \ell_8 s_2 + (-s_1)\overline{s_1 x}).$$

First half: $-\bar{\ell_8}\,(-s_1) = \bar{\ell_8}\,s_1 = -\ell_8 s_1 = s_1\ell_8$ by L3. $s_2(s_1 x) = -(s_1\ell_8)$ by the sed-CD walk in $p \cdot v_1$. Hence first half $= -(s_1\ell_8) + (s_1\ell_8) = 0$. *Second half:* $\overline{s_1 x} = -(s_1 x)$, so $(-s_1)\overline{s_1 x} = s_1(s_1 x) = -x$ by O-LA + N-1 (as in $p \cdot v_1$). $\ell_8 s_2 = x$ by sed-CD (as in $p \cdot v_1$). Hence second half $= x - x = 0$. Combining, $L_{\ell_{16}\cdot p} \cdot v_1 = 0$. ✓

$k = 2, 3$: $L_{\ell_{16}\cdot p} \cdot v_2 = L_{\ell_{16}\cdot p} \cdot v_3 = 0$. By the Pathion Clifford Triad (Theorem 5.1), $\{L_{\ell_{16}\cdot p}, L_{\ell_{16}}\} = 0$ (the third pair of the universal triad), hence $L_{\ell_{16}\cdot p} L_{\ell_{16}} = -L_{\ell_{16}} L_{\ell_{16}\cdot p}$. Applied to v_1 and v_0:

$$L_{\ell_{16}\cdot p} \cdot v_2 \;=\; L_{\ell_{16}\cdot p} L_{\ell_{16}} v_1 \;=\; -L_{\ell_{16}} L_{\ell_{16}\cdot p} v_1 \;=\; -L_{\ell_{16}} \cdot 0 \;=\; 0,$$

and similarly $L_{\ell_{16}\cdot p} \cdot v_3 = 0$. ✓

In all four cases $L_{\ell_{16}\cdot p} \cdot v_k = 0$, hence by linearity $L_{\ell_{16}\cdot p} \cdot \ker L_p = 0$.

Combining. For $v \in \ker L_p$, by Step 1 $\sigma(v) \in E_{2\|p\|^2}(G_p)$, and by Steps 2 + 3:

$$L_w \cdot \sigma(v) \;=\; \sigma\big(L_{\ell_{16}\cdot p} \cdot v\big) \;=\; \sigma(0) \;=\; 0.$$

Since $\sigma : \ker L_p \to E_{2\|p\|^2}(G_p)$ is a bijection (by Step 1 + $\sigma^2 = \mathrm{id}$), this gives $L_w \cdot E_{2\|p\|^2}(G_p) = 0$. □ □

Remark 7.12 (Range of validity)**.** Step 1 of Lemma 7.11 requires the half-localization (7) for both $\iota_1(s_1)$ and $\iota_2(s_2)$, hence the dyadic hypothesis. At random aphanic, the identity $L_p^2 + L_u^2 = -2\|p\|^2 I$ fails (as catalogued in Remark 7.7), and $\ker L_u$ is no longer contained in any single G_p-eigenspace. Step 3 in the q_universal annihilation $L_{\ell_{16}\cdot p} \cdot \ker L_p = 0$ is itself a canonical-Stratum-II identity: at CD-aligned, the recovered basis (8) does not span $\ker L_p$ (the kernel is 12-dimensional there), and the Step 3 walks need a different basis.

7.7 Block decomposition theorem at canonical Stratum-II

Theorem 7.13 (Block decomposition of L_w at canonical Stratum-II)**.** *At every canonical Stratum-II basepoint, L_w is block-diagonal in the orthogonal G_p-eigenspace decomposition $\mathbb{R}^{32} = E_0(G_p) \oplus E_{\|p\|^2}(G_p) \oplus E_{2\|p\|^2}(G_p)$:*

$$L_w \cdot E_0(G_p) \subseteq E_0(G_p), \qquad L_w \cdot E_{\|p\|^2}(G_p) \subseteq E_{\|p\|^2}(G_p), \qquad L_w \cdot E_{2\|p\|^2}(G_p) \;=\; 0.$$

Consequently $[L_w, P_\lambda] = 0$ for each spectral projector P_λ, and

$$[L_w, G_p] \;=\; \sum_\lambda \lambda\, [L_w, P_\lambda] \;=\; 0.$$

Proof. The kernel inclusion $L_w \cdot E_0(G_p) \subseteq E_0(G_p)$ is Lemma 7.9 (kernel of L_p equals $E_0(G_p)$ formally: $G_p v = 0 \iff -L_p^2 v = 0 \iff L_p v = 0$ for the skew-symmetric L_p, since L_p has the same kernel as $L_p^\top L_p = G_p$). The top-eigenspace annihilation $L_w \cdot E_{2\|p\|^2}(G_p) = 0$ is Lemma 7.11. The middle eigenspace $E_{\|p\|^2}(G_p)$ is the orthogonal complement of $E_0(G_p) \oplus E_{2\|p\|^2}(G_p)$ in the standard inner product on $\mathbb{P} \cong \mathbb{R}^{32}$; since L_w is real and skew-symmetric (as $w \in \mathrm{Im}(\mathbb{P})$, by the formal identity $L_w^\top = L_{\bar w} = -L_w$), L_w preserves orthogonal complements. Therefore $L_w \cdot E_{\|p\|^2}(G_p) \subseteq E_{\|p\|^2}(G_p)$.

Block-diagonality in the orthogonal eigenspace decomposition is equivalent to commuting with each spectral projector. The spectral identity $[L_w, G_p] = \sum_\lambda \lambda\, [L_w, P_\lambda] = 0$ is then immediate. □ □

7.8 The CD-Aligned Closure Lemma

Lemma 7.14 (CD-Aligned Closure Lemma)**.** *At every dyadic CD-aligned pathion-aphanic basepoint $p = \iota_1(s_1) + \iota_2(s_2)$ with $s_2 = s_1 \cdot \ell_8$ and $s_1 = (r, 0)$ for some $r \in \mathrm{Im}(\mathbb{O})$ unit, the operators L_w and L_p on $\mathbb{P}$ satisfy $\{L_w, L_p\} = 0$. Consequently, by the centralization chain (Lemma 4.2),*

$$[L_w, G_p] \;=\; 0$$

at every CD-aligned basepoint.

The proof has five steps. Step 1 is a half-embedded decomposition that splits $\{L_w, L_p\}$ into four pathion-level anti-commutators of the operators A_k, B_k. Step 2 establishes the universal mixed-pair flex vanishing $\{A_k, B_k\} = 0$ for any imaginary $s \in \mathrm{Im}(\mathbb{S})$, eliminating two of the four terms via sedenion flexibility. Step 3 reduces the remaining two terms, $\{A_1, A_2\} + \{B_1, B_2\}$, to a pair of sedenion anti-commutators with an $L \leftrightarrow R$ swap. Step 4 collapses both sedenion anti-commutators to zero at CD-aligned via octonion flexibility on two-element subalgebras. Step 5 combines.

Proof of Lemma 7.14. **Step 1: half-embedded decomposition.** By Definition 7.4, $L_p = A_1 + B_2$ and $L_w = A_2 + B_1$. Expanding:

$$\begin{aligned}\{L_w, L_p\} &= \{A_2 + B_1, A_1 + B_2\} \\ &= \{A_2, A_1\} + \{A_2, B_2\} + \{B_1, A_1\} + \{B_1, B_2\} \\ &= \{A_1, A_2\} + \{A_2, B_2\} + \{A_1, B_1\} + \{B_1, B_2\},\end{aligned}$$

using the symmetry $\{X, Y\} = \{Y, X\}$. The four terms are $\{A_1, A_2\}$, $\{B_1, B_2\}$ (the "diagonal" terms in the (ι_1, ι_2)-grading, where each anti-commutator stays within a single embedding) and $\{A_1, B_1\}$, $\{A_2, B_2\}$ (the "mixed" terms, which cross between embeddings of the same s_k).

Step 2: mixed-pair flex vanishing. We claim that for any imaginary $s \in \mathrm{Im}(\mathbb{S})$,

$$\{L_{\iota_1(s)}, L_{\iota_2(s)}\} \;=\; 0 \quad \text{on } \mathbb{P}. \tag{10}$$

This identity holds at every CD level above sedenion and requires no dyadic, aphanic, or CD-aligned hypothesis.

Proof of (10). For $v = (a, b) \in \mathbb{P}$ with $a, b \in \mathbb{S}$:

$$L_{\iota_1(s)} v \;=\; (sa, bs), \qquad L_{\iota_2(s)} v \;=\; (-\bar{b}s, s\bar{a}).$$

By (4), iterating each composition:

$$\begin{aligned}L_{\iota_1(s)} L_{\iota_2(s)} v &= L_{\iota_1(s)}(-\bar{b}s, s\bar{a}) = (s(-\bar{b}s),\ (s\bar{a})s) \;=\; (-s(\bar{b}s),\ (s\bar{a})s), \\ L_{\iota_2(s)} L_{\iota_1(s)} v &= L_{\iota_2(s)}(sa, bs) = (-\overline{bs}\, s,\ s\,\overline{sa}).\end{aligned}$$

The conjugates simplify via CD-2 + imaginarity of s: $\overline{bs} = \bar{s}\,\bar{b} = -s\bar{b}$, so $-\overline{bs}\, s = (s\bar{b})s$; and $\overline{sa} = \bar{a}\bar{s} = -\bar{a}s$, so $s\overline{sa} = -s(\bar{a}s)$. Substituting:

$$L_{\iota_2(s)} L_{\iota_1(s)} v \;=\; ((s\bar{b})s,\ -s(\bar{a}s)).$$

Adding:

$$\{L_{\iota_1(s)}, L_{\iota_2(s)}\} v \;=\; (-s(\bar{b}s) + (s\bar{b})s,\ (s\bar{a})s - s(\bar{a}s)).$$

Each slot is a sedenion expression of the form $(s\xi)s - s(\xi s)$ or $-s(\xi s) + (s\xi)s$, with $\xi = \bar{b}$ or $\bar{a}$. By sed-flex $a(ba) = (ab)a$ on $\mathbb{S}$ (taking $a = s$, $b = \xi$): $s(\xi s) = (s\xi)s$ in $\mathbb{S}$. Hence both slots vanish. $\{L_{\iota_1(s)}, L_{\iota_2(s)}\} v = 0$ for every $v \in \mathbb{P}$, which is (10).

Applying (10) with $s = s_1$ gives $\{A_1, B_1\} = 0$; with $s = s_2$ gives $\{A_2, B_2\} = 0$. Step 1's expansion collapses to

$$\{L_w, L_p\} \;=\; \{A_1, A_2\} \;+\; \{B_1, B_2\}.$$

Step 3: pathion-to-sedenion reduction. We compute $\{A_1, A_2\}$ and $\{B_1, B_2\}$ explicitly on $v = (a, b) \in \mathbb{P}$. By (4):

$$\begin{aligned}A_1 A_2 v &= L_{\iota_1(s_1)}(s_2 a, b s_2) = (s_1(s_2 a),\ (b s_2) s_1), \\ A_2 A_1 v &= L_{\iota_1(s_2)}(s_1 a, b s_1) = (s_2(s_1 a),\ (b s_1) s_2),\end{aligned}$$

hence

$$\{A_1, A_2\}v \;=\; \big(\{L_{s_1}, L_{s_2}\}_{\mathbb{S}}\, a,\; \{R_{s_1}, R_{s_2}\}_{\mathbb{S}}\, b\big), \tag{11}$$

where $\{L_{s_1}, L_{s_2}\}_{\mathbb{S}}\, a \;=\; s_1(s_2 a) + s_2(s_1 a)$ and $\{R_{s_1}, R_{s_2}\}_{\mathbb{S}}\, b \;=\; (bs_2)s_1 + (bs_1)s_2$ are sedenion anti-commutators of left-/right-multiplication operators on $\mathbb{S}$. Similarly:

$$\begin{aligned} B_1B_2v &= L_{\iota_2(s_1)}(-\bar{b}s_2,\; s_2\bar{a}) = (-\overline{s_2\bar{a}}\, s_1,\; s_1\, \overline{-\bar{b}s_2}) \\ &= ((as_2)s_1,\; s_1(s_2 b)), \end{aligned}$$

using $\overline{s_2\bar{a}} = a\bar{s_2} = -as_2$ (so $-\overline{s_2\bar{a}}\, s_1 = (as_2)s_1$) and $\overline{-\bar{b}s_2} = -(\bar{s_2}b) = s_2 b$ (so $s_1\overline{-\bar{b}s_2} = s_1(s_2 b)$). The same computation with $s_1 \leftrightarrow s_2$ gives $B_2B_1v = ((as_1)s_2,\; s_2(s_1 b))$. Therefore

$$\{B_1, B_2\}v \;=\; \big(\{R_{s_1}, R_{s_2}\}_{\mathbb{S}}\, a,\; \{L_{s_1}, L_{s_2}\}_{\mathbb{S}}\, b\big). \tag{12}$$

Note the $L \leftrightarrow R$ swap between (11) and (12): in $\{A_1, A_2\}$ the first slot uses $\{L\}_{\mathbb{S}}$ and the second uses $\{R\}_{\mathbb{S}}$, while in $\{B_1, B_2\}$ the first slot uses $\{R\}_{\mathbb{S}}$ and the second uses $\{L\}_{\mathbb{S}}$.

Combining (11) and (12):

$$\{L_w, L_p\}\, v \;=\; \big((\{L_{s_1}, L_{s_2}\}_{\mathbb{S}} + \{R_{s_1}, R_{s_2}\}_{\mathbb{S}})\, a,\; (\{L_{s_1}, L_{s_2}\}_{\mathbb{S}} + \{R_{s_1}, R_{s_2}\}_{\mathbb{S}})\, b\big). \tag{13}$$

Hence

$$\{L_w, L_p\} \;=\; 0 \text{ on } \mathbb{P} \quad\Longleftrightarrow\quad \{L_{s_1}, L_{s_2}\}_{\mathbb{S}} \;+\; \{R_{s_1}, R_{s_2}\}_{\mathbb{S}} \;=\; 0 \text{ on } \mathbb{S},$$

reducing the pathion-level anti-commutation to a single sedenion-level identity in the sum of left and right anti-commutators of s_1 and s_2.

Step 4: CD-aligned sedenion vanishing. We work at CD-aligned with $s_1 = (r, 0)$ and $s_2 = s_1 \cdot \ell_8 = (0, r)$ (same $r \in \mathrm{Im}(\mathbb{O})$ unit, by L1 / L2 / L4 of Lemma 7.5 specialized to $x' = r$). For a generic $c = (c_1, c_2) \in \mathbb{S}$ with $c_1, c_2 \in \mathbb{O}$ (not necessarily imaginary), we compute the four sed-CD products via (4):

$$\begin{aligned} s_1(s_2 c) = (r,0)(-\bar{c_2}r,\; r\bar{c_1}) \;&=\; \big(-r(\bar{c_2}r),\; (r\bar{c_1})r\big), \\ s_2(s_1 c) = (0,r)(rc_1,\; c_2 r) \;&=\; \big((r\bar{c_2})r,\; -r(\bar{c_1}r)\big), \end{aligned}$$

using sed-CD-1 with the relevant first/second slots and the conjugation identity $\bar{r} = -r$. Adding:

$$\{L_{s_1}, L_{s_2}\}_{\mathbb{S}}\, c \;=\; \big(-r(\bar{c_2}r) + (r\bar{c_2})r,\; (r\bar{c_1})r - r(\bar{c_1}r)\big).$$

Each slot is an octonion expression of the form $r(\xi r) - (r\xi)r$ or its negative, with $\xi \in \{\bar{c_1}, \bar{c_2}\} \subset \mathbb{O}$. By Artin's theorem on the two-generator subalgebra $\langle r, \xi\rangle \subset \mathbb{O}$ (which is associative), oct-flex $r(\xi r) = (r\xi)r$ holds. Hence both slots vanish, and $\{L_{s_1}, L_{s_2}\}_{\mathbb{S}}\, c = 0$ for every $c \in \mathbb{S}$, i.e.,

$$\{L_{s_1}, L_{s_2}\}_{\mathbb{S}} \;=\; 0 \quad \text{at CD-aligned.} \tag{14}$$

The symmetric right-multiplication anti-commutator $\{R_{s_1}, R_{s_2}\}_{\mathbb{S}}\, c = (cs_1)s_2 + (cs_2)s_1$ admits the parallel computation. By sed-CD-1:

$$\begin{aligned} cs_1 &= (c_1, c_2)(r, 0) = (c_1 r,\; -c_2 r), \\ cs_2 &= (c_1, c_2)(0, r) = (rc_2,\; rc_1), \end{aligned}$$

using $\bar{0} = 0$ on the conjugation factors and $-\bar{r} = r$. Iterating:

$$\begin{aligned} (cs_1)s_2 = (c_1 r,\; -c_2 r)(0, r) \;&=\; \big(-r(c_2 r),\; r(c_1 r)\big), \\ (cs_2)s_1 = (rc_2,\; rc_1)(r, 0) \;&=\; \big((rc_2)r,\; -(rc_1)r\big). \end{aligned}$$

Adding:

$$\{R_{s_1}, R_{s_2}\}_{\mathbb{S}}\, c \;=\; \big(-r(c_2 r) + (rc_2)r,\; r(c_1 r) - (rc_1)r\big).$$

By oct-flex on $\langle r, c_k \rangle \subset \mathbb{O}$, both slots vanish. Hence

$$\{R_{s_1}, R_{s_2}\}_{\mathbb{S}} \;=\; 0 \quad \text{at CD-aligned.} \tag{15}$$

Step 5: combine. Substituting (14) and (15) into (13):

$$\{L_w, L_p\}\, v \;=\; \big(0 \cdot a,\; 0 \cdot b\big) \;=\; 0 \in \mathbb{P},$$

for every $v \in \mathbb{P}$. Hence $\{L_w, L_p\} = 0$.

By the centralization chain (Lemma 4.2), $\{L_w, L_p\} = 0 \Longrightarrow [L_w, L_p^2] = 0 \Longleftrightarrow [L_w, G_p] = 0$. □ □

Remark 7.15 (Shortcut via the Δ identity)**.** The Sedenion Δ Identity (Theorem 5.4) states $\Delta(s_1, s_2) := \{L_{s_1}, L_{s_2}\}_{\mathbb{S}} - \{R_{s_1}, R_{s_2}\}_{\mathbb{S}} = 0$ for all imaginary $s_1, s_2 \in \mathrm{Im}(\mathbb{S})$, with no orthogonality or CD-alignment hypothesis. Hence $\{L_{s_1}, L_{s_2}\}_{\mathbb{S}} = \{R_{s_1}, R_{s_2}\}_{\mathbb{S}}$ universally, and either of (14) / (15) implies the other. The direct sed-CD walks above are self-contained and avoid invoking Δ, but the shortcut is a cleaner statement when integrating into v2.

Remark 7.16 (Sharpness of CD-aligned)**.** The flex argument of Step 4 fails at canonical Stratum-II. With $s_1 = (r, 0)$ and $s_2 = (0, x')$, $x' \neq r$ in $\mathrm{Im}(\mathbb{O})$ orthogonal, the sed-CD walks of Step 4 produce cross-products $r(\bar{c_k}x')$ and $(r\bar{c_k})x'$ rather than the symmetric form $r(\xi r) = (r\xi)r$ used at CD-aligned. The associator $[r, \bar{c_k}, x']_{\mathbb{O}}$ is generally nonzero in $\mathbb{O}$ — the three-element subalgebra $\langle r, x', \bar{c_k} \rangle$ is non-associative for generic $\bar{c_k}$ — and the flex collapse fails. Diagnostic v5 records $\|\{L_w, L_p\}\|_F = 16$ across the sampled canonical Stratum-II family, consistent with this sharpness.

7.9 Master theorem and consequences

The three lemmas combine into a single master statement governing the dyadic q_antisym centralization.

Theorem 7.17 (Two-Regime Dyadic Closure)**.** *Let $p \in \mathcal{P}_{\mathbb{P}}^{\mathrm{aph}}$ be a dyadic basepoint (Definition 7.1) at which* $\mathrm{Spec}(G_p)$ *is three-valued. Then for every $a \in \mathfrak{q}_p^{\mathrm{anti}} = \mathrm{span}_{\mathbb{R}}\{u, w\}$,*

$$[L_a, G_p] \;=\; 0.$$

The closure mechanism splits along the dyadic family:

(1) *(u-side, all dyadic) For every dyadic basepoint, $\{L_u, L_p\} = 0$, hence $[L_u, G_p] = 0$ via the centralization chain (Lemma 4.2). [Lemma 7.6]*

(2) *(w-side, canonical Stratum-II) At every canonical Stratum-II basepoint, L_w is block-diagonal in the G_p-eigenspace decomposition $E_0 \oplus E_{\|p\|^2} \oplus E_{2\|p\|^2}$ (with kernel preservation and top-eigenspace annihilation), hence $[L_w, G_p] = 0$. [Theorem 7.13]*

(3) *(w-side, CD-aligned) At every CD-aligned basepoint, $\{L_w, L_p\} = 0$, hence $[L_w, G_p] = 0$ via the centralization chain. [Lemma 7.14]*

By linearity of $[\cdot, G_p]$ in the first argument, every $a \in \mathfrak{q}_p^{\mathrm{anti}}$ satisfies $[L_a, G_p] = 0$ at every dyadic basepoint (canonical Stratum-II via (1) + (2); CD-aligned via (1) + (3)).

Proof. Direct combination of the three cited lemmas. The dyadic family partitions into the canonical Stratum-II subfamily and the CD-aligned subfamily (Definition 7.2). At canonical Stratum-II, items (1) and (2) supply the u- and w-side closures respectively; at CD-aligned, items (1) and (3) supply them. Linearity in $a \in \mathfrak{q}_p^{\mathrm{anti}}$ extends the basis closures $[L_u, G_p] = [L_w, G_p] = 0$ to all of $\mathfrak{q}_p^{\mathrm{anti}} = \mathrm{span}_{\mathbb{R}}\{u, w\}$. □ □

Corollary 7.18 (Algebraic upgrade of Theorem 4.3 sufficiency)**.** *The sufficiency direction* $\mathfrak{q}_p^{(32)} \subseteq \mathcal{K}(p)$ *in Theorem 4.3 (Pathion Q-Centralizer Vanishing at Dyadic Basepoint) admits an algebraic proof at every dyadic basepoint with three-valued* $\mathrm{Spec}(G_p)$*, via the centralization chain Lemma 4.2 and:*

- $[L_a, G_p] = 0$ *for* $a \in \mathfrak{q}_p^{\mathrm{univ}}$*, by the Pathion Clifford Triad Theorem (5.1) and the centralization chain;*
- $[L_a, G_p] = 0$ *for* $a \in \mathfrak{q}_p^{\mathrm{anti}}$*, by Theorem 7.17 above.*

Together these give $[L_a, G_p] = 0$ *for every* $a \in \mathfrak{q}_p^{(32)} = \mathfrak{q}_p^{\mathrm{univ}} \oplus \mathfrak{q}_p^{\mathrm{anti}}$*, hence* $[L_a, \pi_p] = 0$ *and* $\mathcal{C}_{a,b}(p) = 0$ *for all* b*.*

Proof. Both bullet points are established algebraically: the first via Theorem 5.1 (proved fully algebraically in §5 via the chain Lemma A + Δ-reduction + Ω-theorem), the second via Theorem 7.17 above. The centralization chain Lemma 4.2 then propagates $[L_a, G_p] = 0$ to $[L_a, \pi_p] = 0$ to $\mathcal{C}_{a,b}(p) = 0$ for every b. □ □

Remark 7.19 (Resolution of the Q1 mechanism question)**.** Question (Q1) of §8 notes that $[L_a, G_p] = 0$ for $a \in \mathfrak{q}_p^{\mathrm{anti}}$ holds only at three-valued spectrum (not universally), and conjectured that the responsible mechanism is the Lagrange-3 polynomial form $G_p^+ = \frac{7}{4\|p\|^4} G_p - \frac{3}{4\|p\|^6} G_p^2$, valid only at three-valued spectrum.

Theorem 7.17 above shows that the load-bearing algebraic mechanism is *not* the Lagrange-3 polynomial form, but the half-localization identity (7) (powering the u-side closure across all of dyadic), together with one of two regime-specific arguments on the w-side: σ-image identification of $E_{2\|p\|^2}(G_p)$ with $\sigma(\ker L_p)$ at canonical Stratum-II (Lemma 7.11), or octonion flexibility on two-element subalgebras at CD-aligned (Lemma 7.14). The Lagrange-3 identity is consistent with the closure (the G_p^+-form as a polynomial in G_p commutes with anything that commutes with G_p), but is not used in any step of the derivation.

The dyadic restriction of the closure is sharp: Remark 7.7 catalogues the u-side failure at random aphanic via cross-half octonion associators; Remark 7.12 catalogues the w-side failure at random aphanic via the breakdown of $L_p^2 + L_u^2 = -2\|p\|^2 I$; Remark 7.16 catalogues the canonical-Stratum-II failure of the CD-aligned argument via three-element octonion subalgebras. Each of the three lemmas has its own sharpness witness in the diagnostics.

7.10 Cross-reference to the v45-A proof note

For readers tracking the v45-A diagnostic / proof-note sequence (`state/v45a_l_w_spectral_closure.md`), the lemma names used in this section map back as follows:

- Lemma 7.6 ("u-Closure Lemma") ≡ proof-note Lemma A.
- Lemma 7.8 (Recovered Kernel Basis) + Lemma 7.9 (B1 Kernel Preservation) + Lemma 7.11 (B2 Top-Eigenspace Annihilation) + Theorem 7.13 (Block decomposition of L_w) ≡ proof-note Lemma B (with B1 / B2 broken out as separate sub-lemmas).
- Lemma 7.14 ("CD-Aligned Closure Lemma") ≡ proof-note Lemma C.

Lemma A (5.2) is the associator-symmetry identity used in the proof of the Pathion Clifford Triad Theorem; it is unchanged and unrelated to the present Lemma 7.6.

8 Discussion

8.1 Connection to sedenion U(2) holonomy

At sedenion the holonomy group of the kernel bundle has been shown to be $U(2)$ [2]: as a group, $(U(1) \times SU(2))/\mathbb{Z}_2$, incidentally the gauge group of the electroweak interaction, though neither cited paper claims a physical identification. The Pathion Clifford Triad Theorem provides a structural lift of this picture: the three operators $L_p, L_{\ell_{16}}, L_{\ell_{16} \cdot p}$ form an anti-commuting triad whose squares are not scalar but whose anti-commutation pattern mirrors the generators of an $\mathfrak{su}(2)$ Lie algebra under the operator commutator. Whether the kernel bundle at pathion carries an analogous $U(n)$ structure for some n, and whether the antisymmetric Clifford pair $\mathfrak{q}_p^{\text{anti}}$ contributes a second Stiefel–Whitney class, are natural questions pointed at by the present results but not resolved here.

8.2 Why the dimension drops

The most striking phenomenon at pathion is the dimensional drop of the universal kill set from 5 at sedenion to 3 at pathion. We can now explain why: the sedenion kill set $\mathfrak{q}_p^{(16)}$ contains five generators because *all* of them centralize G_p at sedenion, where the spectrum is universally $\{0^4, 2^8, 4^4\}$ and the Lagrange-3 identity holds globally. At pathion the universal spectrum fails (Proposition 3.1), and only the three universal-Clifford generators retain their G_p-centralization. The two extra generators revert to dyadic-only behavior, producing the R136 vs R137 distinction.

In other words, sedenion's dim-5 kill set is over-determined in two distinct ways: (a) its three universal-Clifford generators centralize G_p via the Clifford triad mechanism, valid at every CD level; and (b) its two antisymmetric generators centralize G_p via the Lagrange-3 identity, which is contingent on three-valued spectrum and fails to lift to pathion. Sedenion happens to satisfy both conditions simultaneously; pathion does not.

8.3 Closed in this paper

(Q1, closed) Algebraic centralization $[L_a, G_p] = 0$ for $a \in \mathfrak{q}_p^{\text{anti}}$ at dyadic basepoints. A prior version of this discussion left as an open question whether $[L_a, G_p] = 0$ for $a \in \mathfrak{q}_p^{\text{anti}}$ at three-valued spectrum admits a purely algebraic derivation (the centralization is not universal, since $\{L_a, L_p\}$ fails for mixed pairs across the σ-grading at random aphanic). The Two-Regime Dyadic Closure (Theorem 7.17, Section 7) supplies the derivation. The closure mechanism is two-regime within the dyadic family rather than a single global identity: the u-side closes universally across the dyadic family by half-localization (Lemma 7.6); the w-side closes by block-diagonality of L_w in the G_p-eigenspace decomposition at canonical Stratum-II (Theorem 7.13), or by direct Clifford anti-commutation $\{L_w, L_p\} = 0$ at the CD-aligned subfamily (Lemma 7.14). The Lagrange-3 polynomial form $G_p^+ = \frac{7}{4\|p\|^4} G_p - \frac{3}{4\|p\|^6} G_p^2$ is consistent with the closure but is not the load-bearing element; see Remark 7.19. The dyadic restriction is sharp at three points (Remarks 7.7, 7.12, 7.16).

8.4 Open questions

Two concrete questions remain.

(Q2) Spectrum non-invariance and orbits: substantially closed. The non-invariance of $\text{Spec}(G_p)$ recorded in Proposition 3.1 is now resolved on two of the 3-valued classes:

- *Class A*, $\text{Spec}(G_p) = \{0^{12}, 2^8, 4^{12}\}$, is closed by R138 across the six named dyadic subcases (framework state at v46-C18 close).

- *Class B Stratum II*, $\mathrm{Spec}(G_p) = \{0^4, 2^{24}, 4^4\}$, is closed by R140 (framework state at v47-A22 close) into the three G_2-orbits described in Section 3. An earlier conjecture that the orbit classes were parameterized by an integer counting non-trivial cross-half projections of (s_1, s_2) is *not* the correct invariant; the corrected invariant is the pair (component norm tuple, $\mathrm{cb}^2_{\mathrm{can}}$), with three admissible values $(1, 0, 0, 1, 0)$, $(0, 1, 1, 0, 0)$, and $((1/2)^4, 12)$.

The reduction was not a single argument. R139 (v47-A7) closed the two two-quad layouts (canonical and mirror) as a sufficient-condition theorem. A subsequent layout-exhaustion conjecture ("Reading 3") was refuted by an explicit third G_2-orbit at $p_3 = \frac{1}{\sqrt{2}}(e_1 + e_{25} + e_{10} + e_{18})$, characterized by all four pathion-quad slots having equal magnitude $1/\sqrt{2}$ together with the co-alignment relations $\alpha_1 \parallel \beta_2$ and $\beta_1 \parallel \alpha_2$ in $\mathrm{Im}(\mathbb{O})$ and the sign-correlation parity $\mathrm{sign}(\alpha_1)\mathrm{sign}(\beta_2) = \mathrm{sign}(\beta_1)\mathrm{sign}(\alpha_2)$. R140 replaces the refuted two-layout exhaustion with the corrected three-orbit classification and, in particular, formally classifies p_3 as the third orbit. The third-orbit closed-form anatomy on the two-parameter family $p(a, b) = a(e_1 + e_{25}) + b(e_{10} + e_{18})$ (commutator-balance quartics $\mathrm{cb}^2_{\mathrm{can}}$, $\mathrm{cb}^2_{\mathrm{mir}}$, $\mathrm{cb}^2_{\mathrm{sed}}$; the canonical p_3 rank-6 operator skeleton $ab \cdot (M \oplus N^\top)$; and a rank obstruction to any rank-preserving σ-bridge between the canonical and mirror partitions) is recorded in Section 3 and was derived in framework state at v47-A24 through v47-A29. The generic eight-valued class C ($\rho = 4$, eight-valued Gram spectrum) remains open as an orbit classification, but its natural 10-tuple of continuous G_2-invariants is locally/generically *non-separating* (R142, Section 3): on a regular Class C neighborhood the orbit space has dimension $13 = 27 - 14$ while the tuple maps into $\mathbb{R}^{10}$, a separation deficiency of exactly 3. The central-inversion collision witness and the local 13-tuple completion (R143) are recorded in Section 3; what remains open is a continuous, non-inversion finite-distance collision family and a *global* orbit classification. The $\rho = 3$ single-component-redundancy classes D, E, F are closed by R141 (Section 3): the cross-quad trace fingerprint $\mathrm{tr}(M) \in \{6, 0\}$ separates D from E/F, and within Class D the singular-value tuple $\varsigma(M)$ further stratifies into Fano-triple (42 members, golden-ratio singular values) and non-Fano (168 members) G_2-orbits. Neither a continuous ς-classification of the full $\rho = 3$ locus nor a representation-theoretic explanation of the Fano-triple golden-ratio phenomenon is yet available.

(Q3) Higher levels of the tower. The chingon algebra ($n = 64$) is the next Cayley–Dickson level. The techniques of this paper extend in principle to chingon, but the spectrum non-invariance phenomenon may sharpen further. Whether the universal Clifford triad survives, whether the R140 three-orbit structure persists, whether the R141 cross-quad trace fingerprint and ς-stratification extend, and whether a new structural break appears at $n = 64$, are open.

A Computational certification

All numerical claims in this paper are verified at machine precision (IEEE 754 double, residual $\leq 10^{-12}$ unless noted) by self-contained C++ probes. The $\Gamma_{\mathcal{C}}$ nullity certifications cited in the necessity directions of Theorems 4.1 and 4.3 are reported in those proofs with their sample descriptions; their run logs are part of the framework record. The probes use a recursive Cayley–Dickson multiplication engine with no external dependencies; full source code is available from the author. For each result we report the maximum residual observed across the indicated sample.

Identity	**Sample**
Pathion Clifford Triad: $\{L_a, L_b\} = 0$ for $a, b \in \mathfrak{q}_p^{\text{univ}}$	$N = 4$ random aphanic
Sedenion Δ identity, orthogonal imaginary	$N = 32$ samples
Sedenion Δ identity, non-orthogonal imaginary	$N = 32$ samples
$\Delta((e_1, 0), (0, e_2))$ at canonical dyadic case	exact
4-block decomposition cases (i), (ii), (iv)	$N = 32$ each
$\Omega(A, B; c) = 0$, generic $A, B, c \in \mathbb{O}$	$N = 32$ samples
$\Omega(A, B; c) = 0$, $A, B \in \mathrm{Im}(\mathbb{O})$	$N = 32$ samples
Ω diagonal: $\bar{A}(Ac) - (c\bar{A})A$ via Artin	$N = 64$ samples
$\mathbb{Z}_2$ split: $\{L_a, L_b\}$ within $\mathfrak{q}_p^{\text{univ}}$	$N = 8$ random aphanic
$\mathbb{Z}_2$ split: $\{L_a, L_b\}$ within $\mathfrak{q}_p^{\text{anti}}$	$N = 8$ random aphanic
$\mathbb{Z}_2$ split: $\{L_a, L_b\}$ mixed (separation)	$N = 8$ random aphanic
σ algebra automorphism: $\sigma(xy) = \sigma(x)\sigma(y)$	$N = 32$ random pathions
Operator transport: $L_{\sigma(a)} - \sigma L_a \sigma^{-1}$	$N = 16$ random pathions
Transport residual: $\{L_u, L_w\} - \sigma\{L_p, L_{\ell_{16}\cdot p}\}\sigma^{-1}$	$N = 8$ random aphanic
R141: Class D/E/F cross-quad fingerprint (v47-B7a through B9)	
R141 block form $[2I_8, M; M^\top, 2I_8]$ at canonical D	long-double reconstruction
R141 $\mathrm{tr}(M) = 6$ at canonical D	bit-exact
R141 $\mathrm{tr}(M) = 0$ at canonical E and F	bit-exact
R141 $\varsigma(M)$ Fano orbit: $(0, 1/\varphi, 1^5, \varphi)$	42 D-variants
R141 $\varsigma(M)$ non-Fano orbit: $(0^2, 1/\varphi, 1^3, \sqrt{2}, \varphi)$	168 D-variants
R141 G_2-equivariance: $\mathrm{tr}(M)$, $\|M\|_F^2$, $\varsigma(M)$ preserved	18 G_2 elements
R141 V_1 split: $\dim(V_1 \cap U_\pm^{\text{can}}) = 4$	canonical D
*R142: Class C 10-tuple incompleteness, **exact** single-point witnesses (v47-C7b/C8, exact rational arithmetic*	
R142 (L1) constraint minor 3×3, $\det = 80 \neq 0$ ($\dim M = 27$)	exact, one witness
R142 (L2) orbit minor $14 \times 14 \neq 0$ ($d_O = 14$)	exact, one witness
R142 (L3) projected-Jacobian minor $10 \times 10 \neq 0$ ($\mathrm{rank}\, dF|_{TM} = 10$)	exact, one witness
R142 (L3) Jacobian cross-check (truncated-poly), 13×13 integer minor	300/300 entries
R142 (K1) $dF \cdot \mathrm{Orb}^\top = 0$ (orbit $\subseteq \ker dF$)	exact, one witness
R142 (K2) $C \cdot \mathrm{Orb}^\top = 0$ (orbit tangent to M)	exact, one witness

Build environment. GCC 13.3.0 / Clang 17 / MSVC 2022, all targets C++17 with no external dependencies. Hardware: AMD Ryzen 9 7950X3D. All probes compile and run in under one second on a single core; wall-clock times reported in the source-level CI output bundled with the project repository.

References

[1] K. Skiouris. Algebraic Hiddenness in the Sedenion Algebra: Characterization, Spectral Structure, and Bundle Topology. Zenodo, doi:10.5281/zenodo.22016333, 2026.

[2] K. Skiouris. Aphanics and Creative Curvature: A Geometric Theory of Hidden Generation in Dialectical Cognitive Systems. Zenodo, doi:10.5281/zenodo.22032281, 2026.